

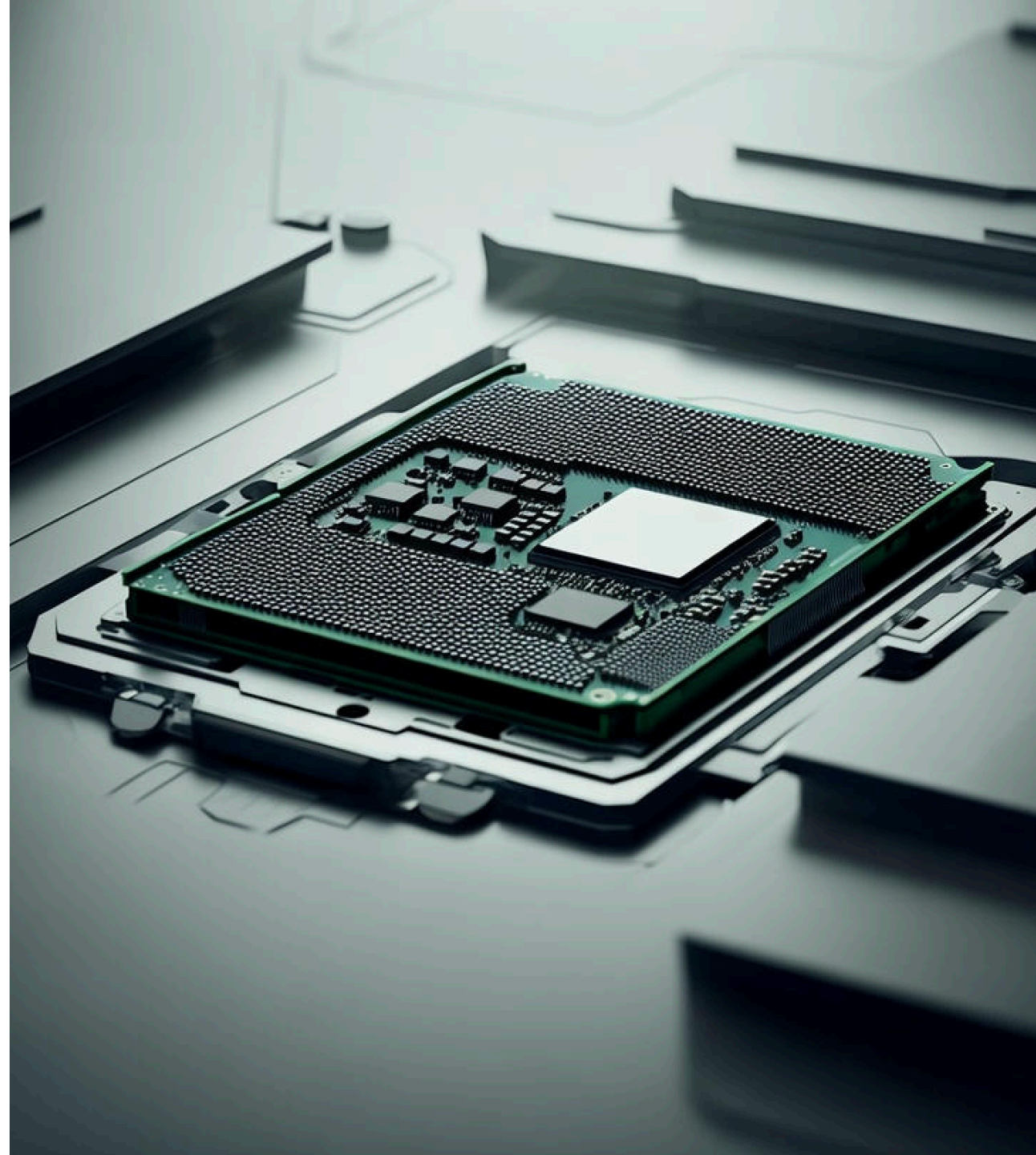
Make Your Code Faster with Data Parallelism

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X: [@lemire](#)

GitHub: <https://github.com/lemire/>



Where I am coming from

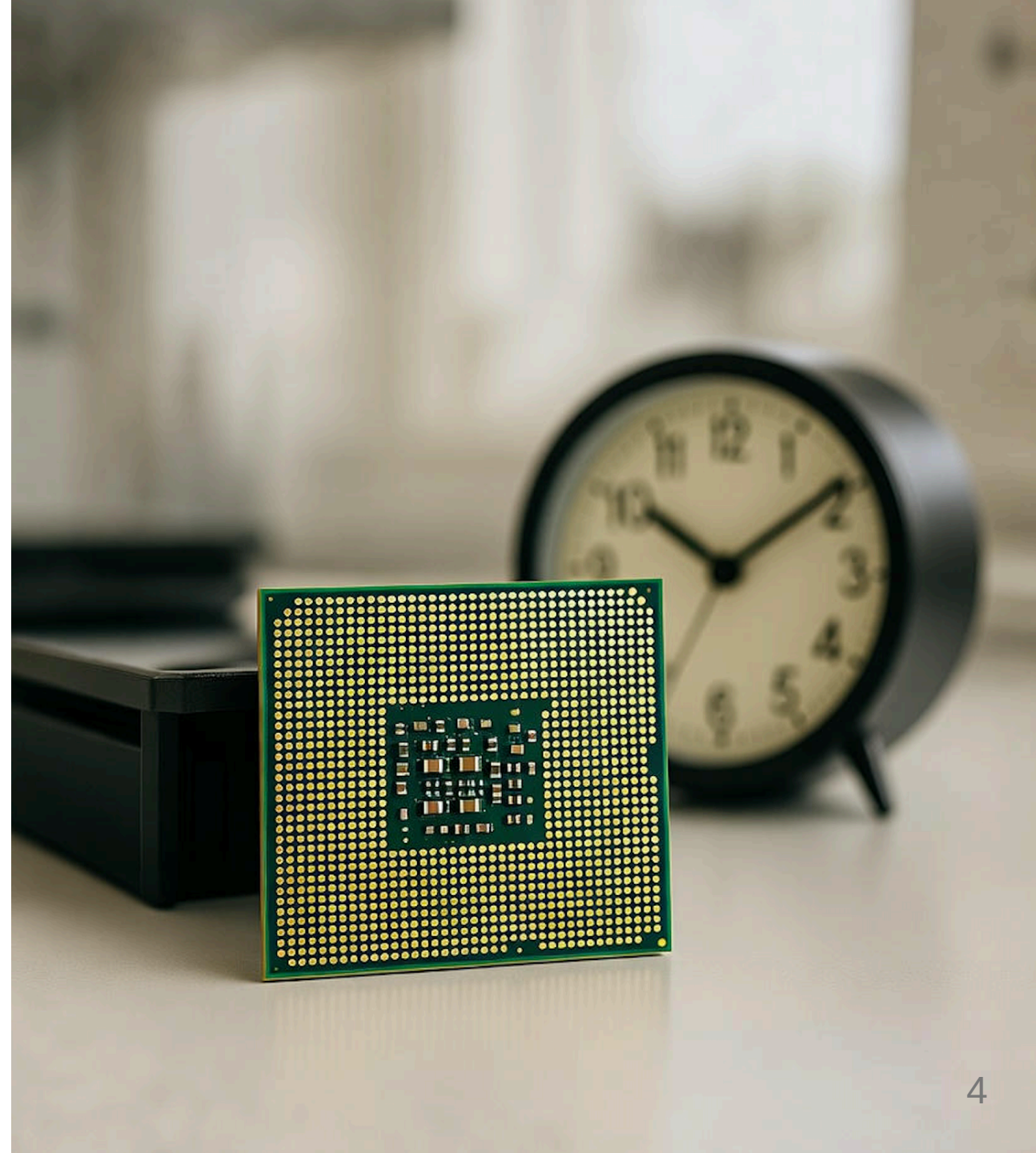
- Author of libraries you are probably running right now: **simdjson**, **simdutf**, **fast_float**, **Roaring Bitmaps**.
- Shipped inside .NET, Rust, Go, Node.js, Bun, Deno, Google Chrome, Safari, ClickHouse.
- Node.js core contributor (C++ review, performance, security).
- Editor, *Software: Practice and Experience* (Wiley, founded 1971).

Part 1

Parallelism for the win

Some numbers

- Time is discrete: the clock cycle
- Processors: ~ 4 GHz
- One cycle is 0.25 nanoseconds
- Light travels 7.5 cm per cycle
- **One byte per cycle is only 4 GB/s**



Frequencies and transistors

processor	year	frequency	transistors
Pentium 4	2000	3.8 GHz	0.040 billions
Intel Haswell	2013	4.4 GHz	1.4 billions
Apple M1	2020	3.2 GHz	16 billions
Apple M3	2024	4.05 GHz	25 billions
Apple M4	2024	4.5 GHz	28 billions
AMD Zen 5	2024	5.7 GHz	50 billions

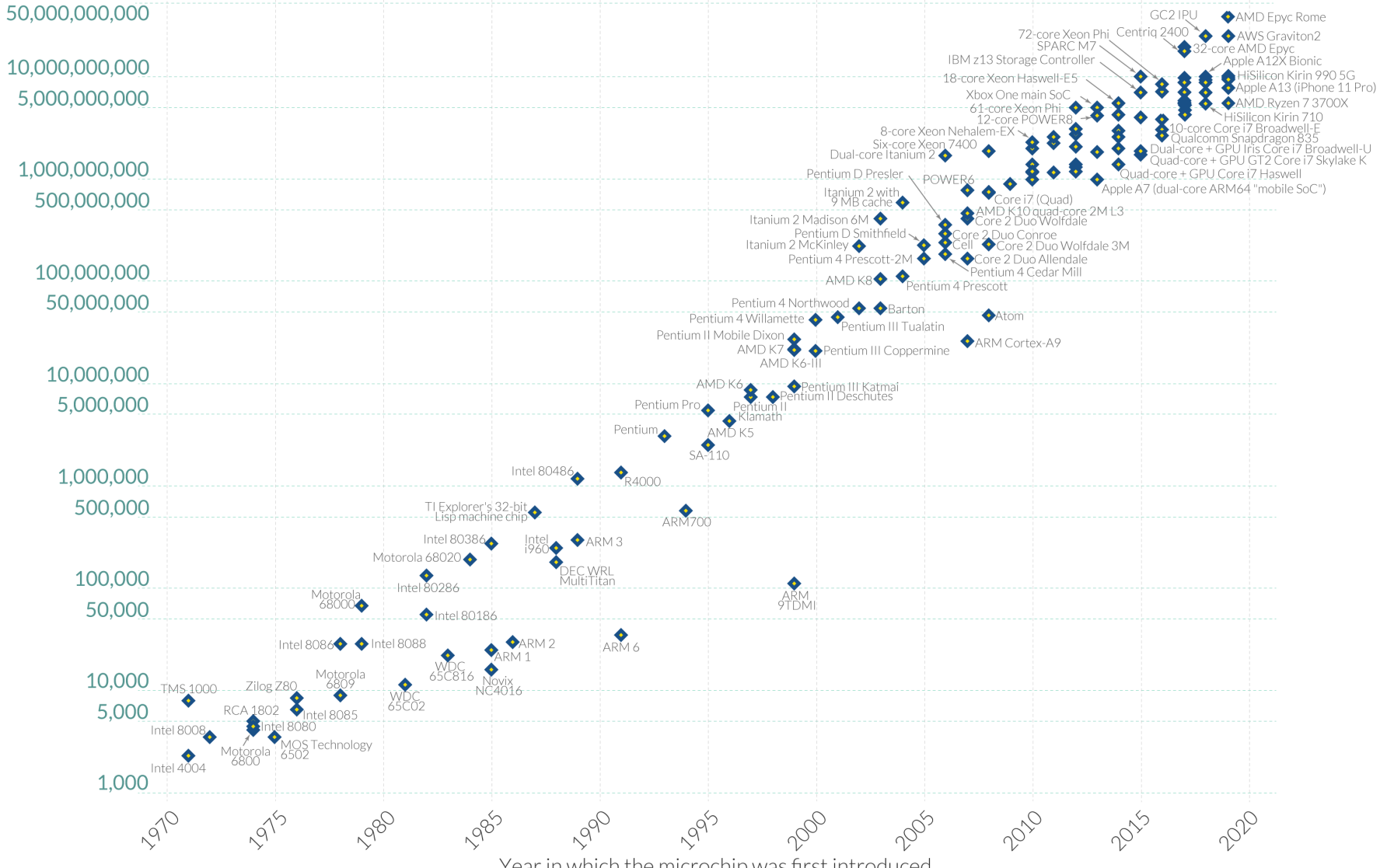
Frequency: +50% in 25 years. Transistors: 1000x.

Moore's Law: The number of transistors on microchips has doubled every two years



Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Transistor count



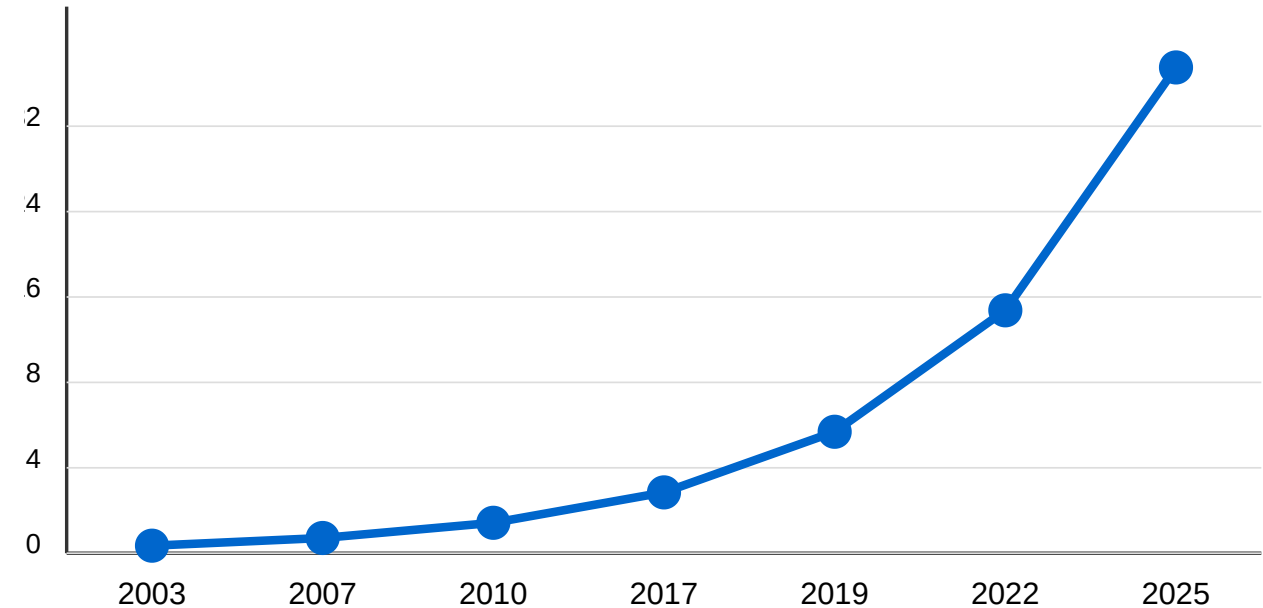
Where do the transistors go?

- More cores
- More superscalar execution (more instructions per cycle)
- Better speculative execution (→ more instructions per cycle)
- More cache, more memory-level parallelism (→ more instructions per cycle)
- Better **data-level parallelism** (SIMD) (→ **fewer instructions**)

The last one is different: the hardware does not do it *for* you.

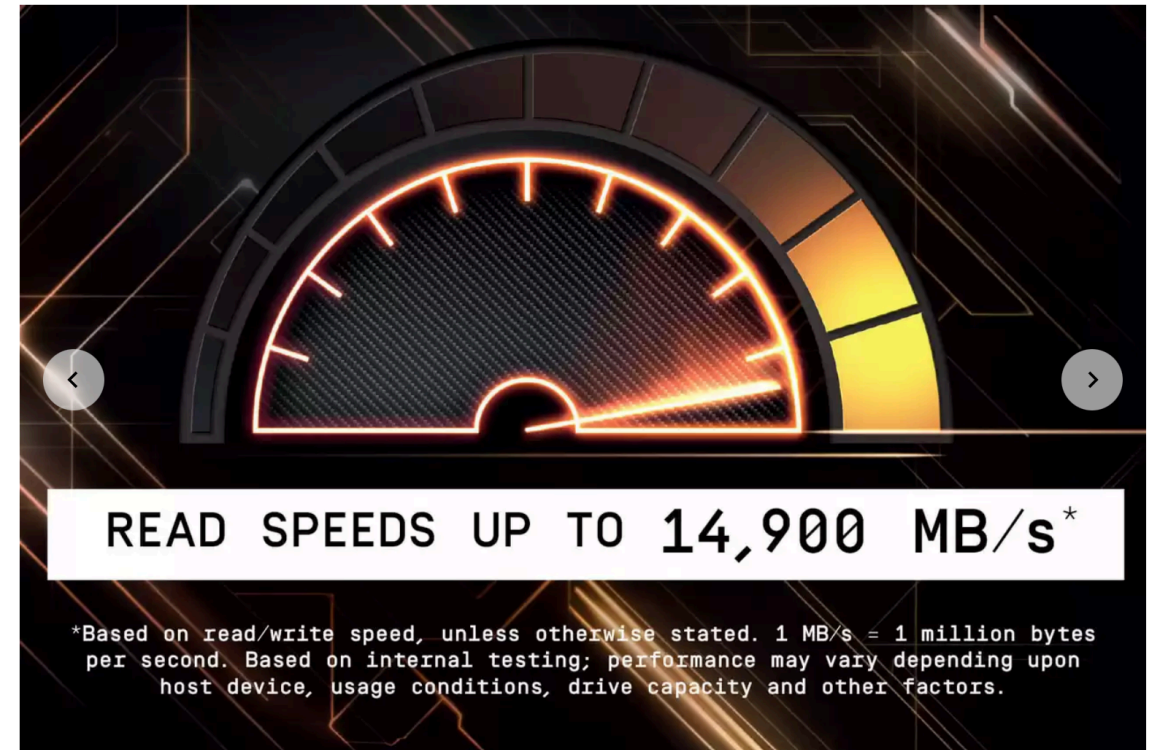
Doubles every 3 years

Specification	year	one channel
PCIe 1.x	2003	500 MB/s
PCIe 2.x	2007	1 GB/s
PCIe 3.x	2010	2 GB/s
PCIe 4.x	2017	4 GB/s
PCIe 5.x	2019	8 GB/s
PCIe 6.x	2022	16 GB/s
PCIe 7.x	2025	32 GB/s



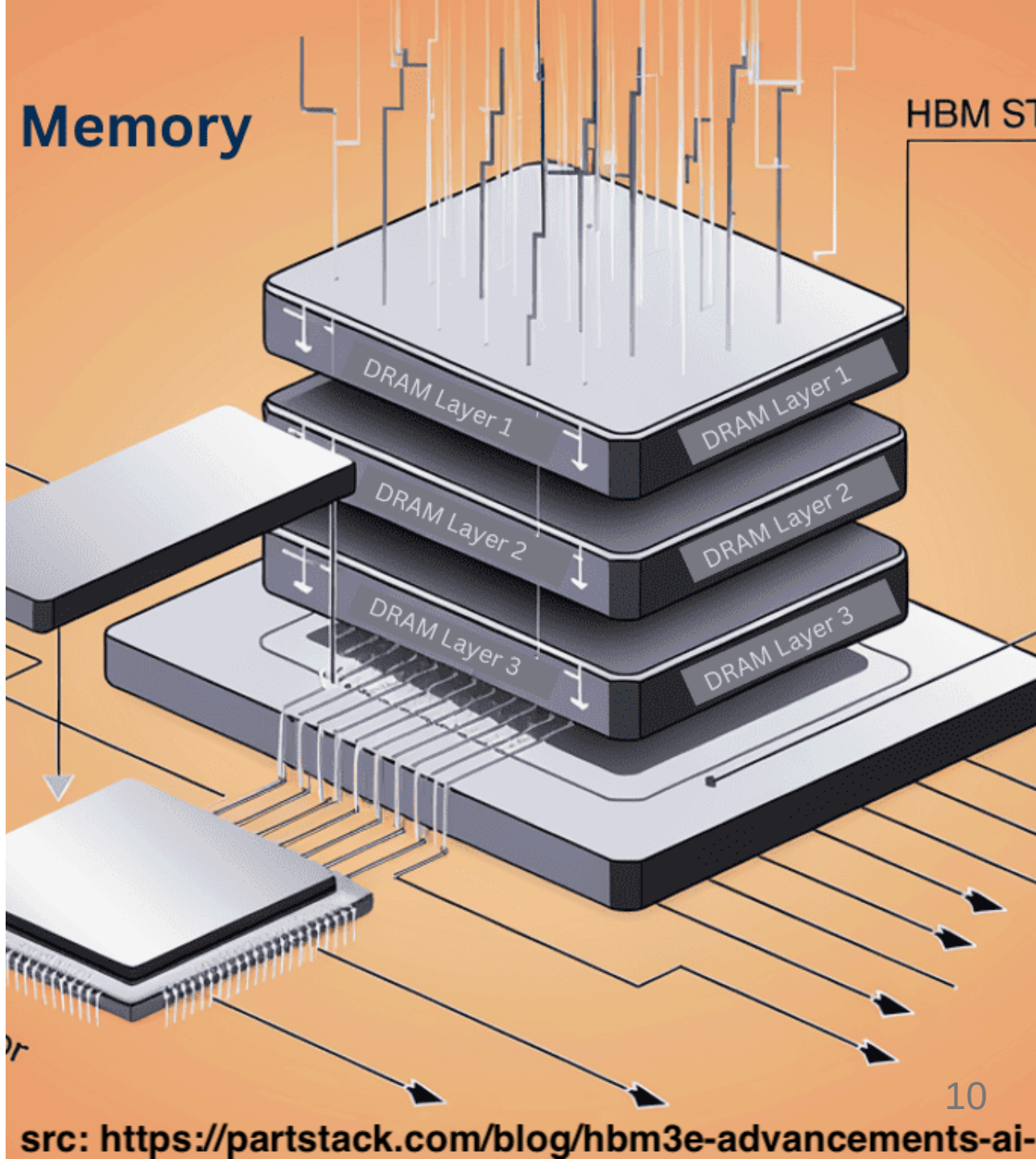
Disk at gigabytes per second

- Sony PlayStation 5 (2020):
5 GB/s
- Sony PlayStation 6 (2027):
15 GB/s (?)



High Bandwidth Memory

Generation	Year	Bandwidth (per stack)
HBM2E	2020	~460 GB/s
HBM3	2022	819 GB/s
HBM3E	2024	~1.2 TB/s
HBM4	2026	>2.8 TB/s
HBM4E	2027	~4 TB/s



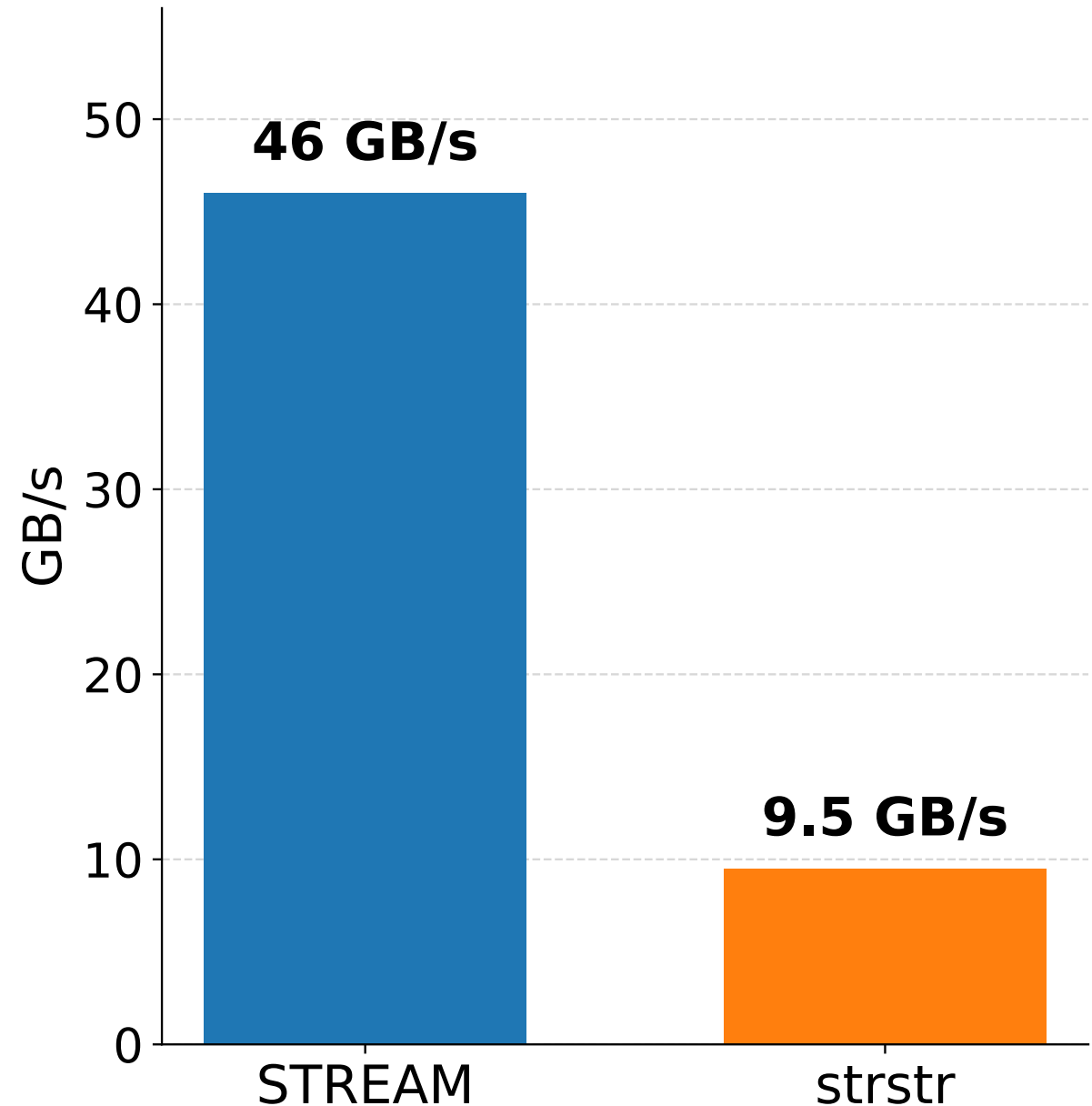
The squeeze

- Your storage delivers gigabytes per second.
- Your memory delivers hundreds of gigabytes per second.
- Your software processes bytes **one at a time**.

You are CPU bound.

Zen 5 AWS (EPYC 9R45, c8a)

- STREAM single thread bandwidth: 46 GB/s
- `strstr`, 32-byte needle: 9.5 GB/s



Part 2

The kinds of parallelism

1. Multicore

- The one everyone knows.
- Hard: synchronization, false sharing, non-determinism.
- Does nothing for a single request that must finish in 2 ms.
- Multiplies your energy bill by the number of cores.

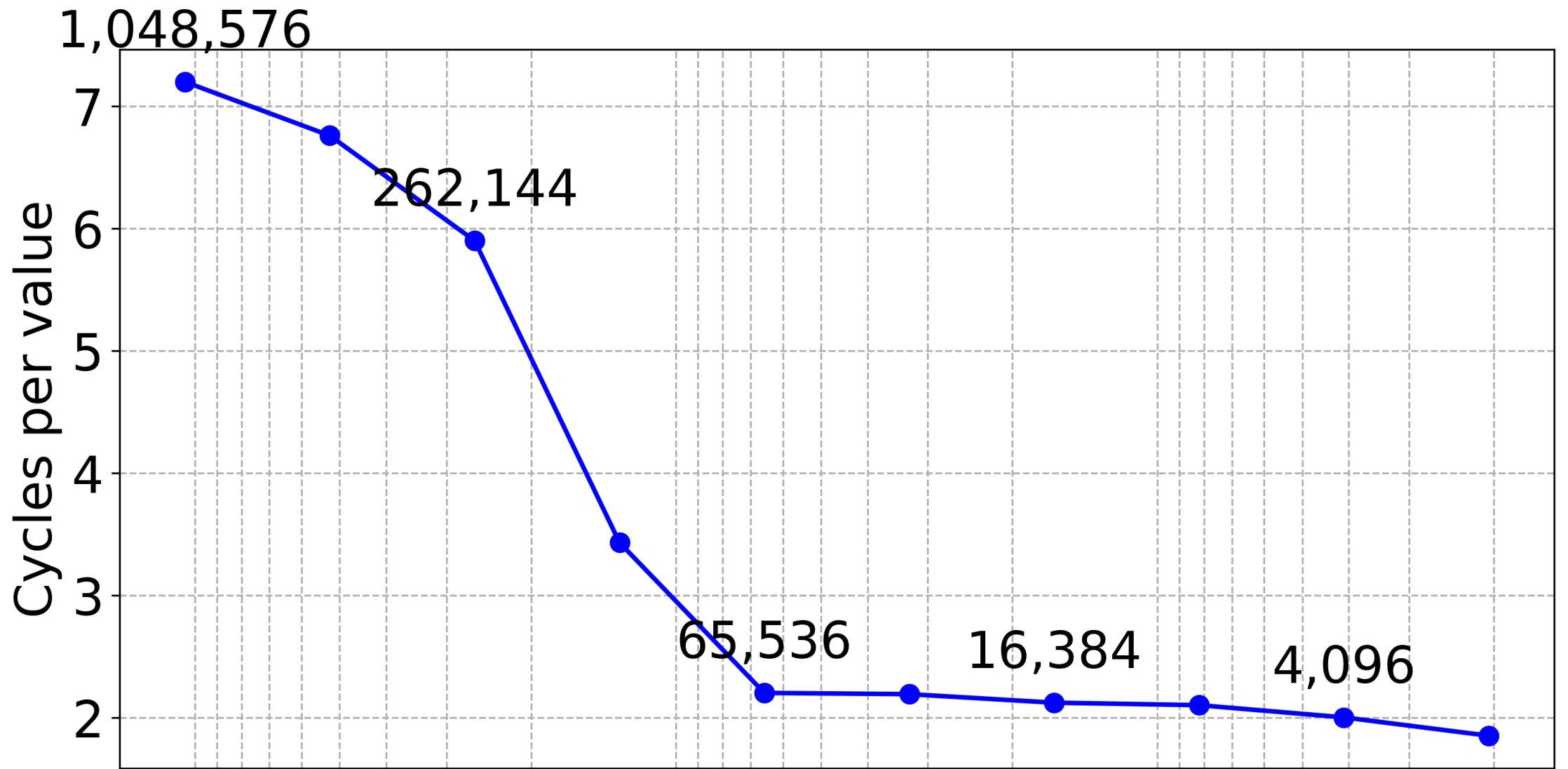
Do this last, not first.

2. Superscalar and speculative execution

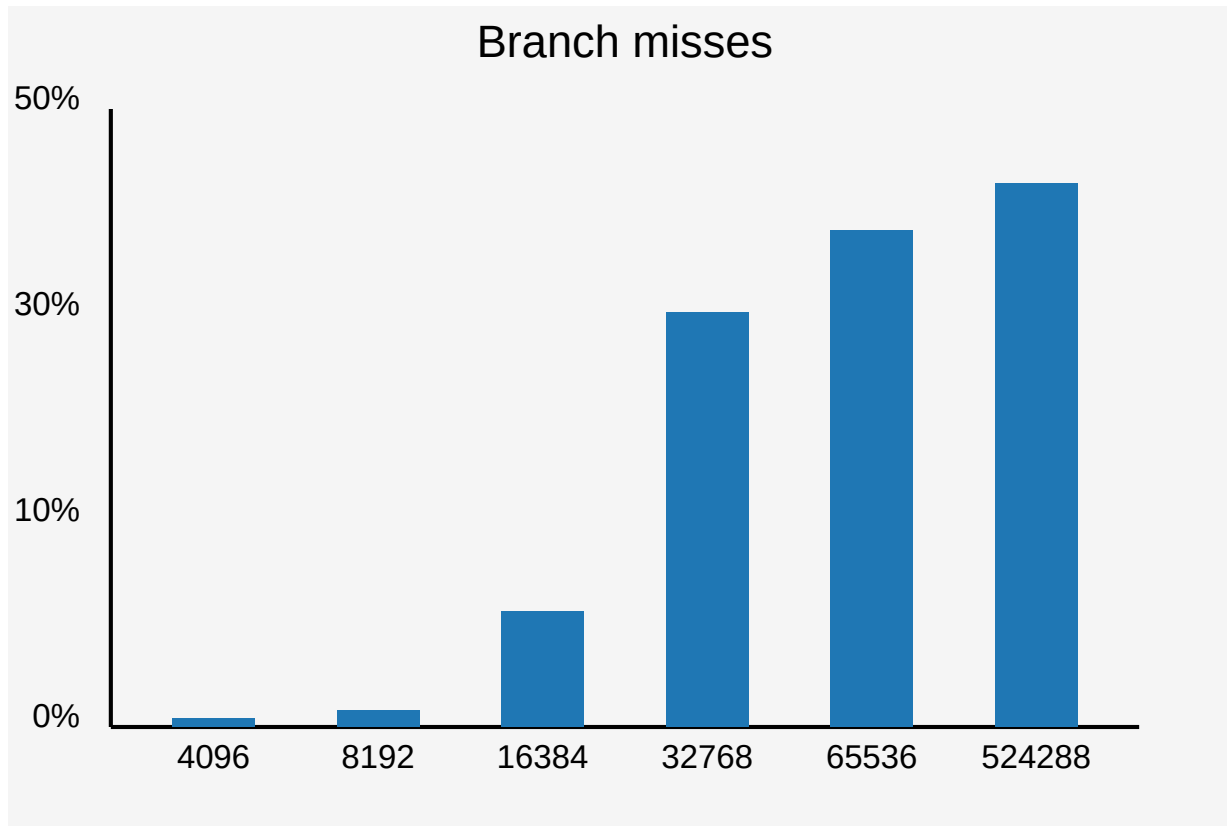
processor	year	arithmetic logic units	SIMD units
Pentium 4	2000	2	2×128
AMD Zen 2	2019	4	2×256
Apple M*	2019	6+	4×128
Intel Lion Cove	2024	6	4×256
AMD Zen 5	2024	6	4×512

Processors *predict* branches and execute code *speculatively*. A misprediction costs 10–20 cycles.

How much can your processor learn?



Apple M4 can learn 10,000 random (0/1) branches



1. Benchmark over **massive** inputs, or you are measuring the branch predictor's memory.
2. Prefer a branchless solution when it costs you nothing.

3. Memory-level parallelism

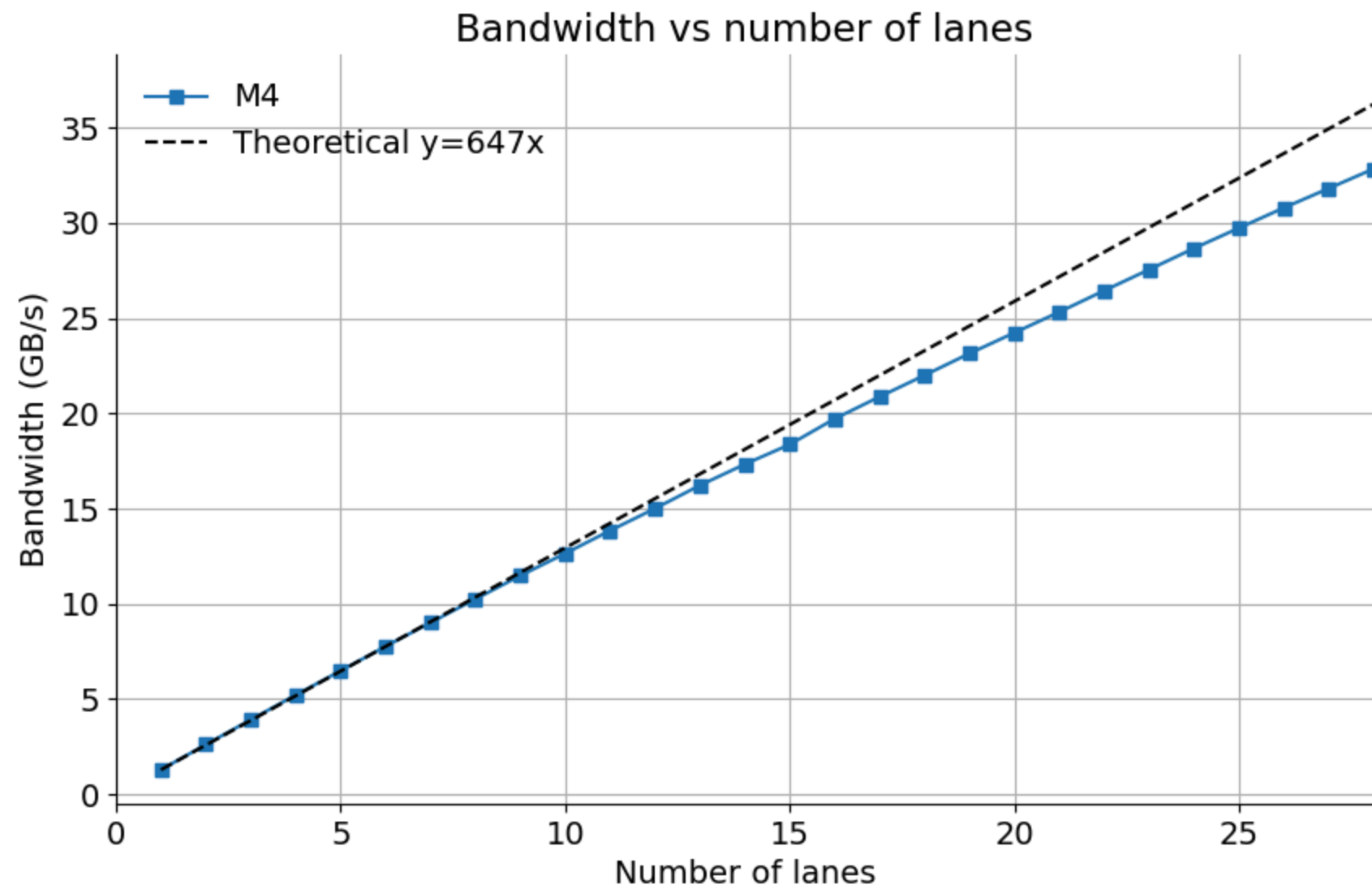
- Latency to RAM: ~100 ns (400 cycles).
- But you can have dozens of loads in flight at once.

cycle	action	action	pizza en route
1	order pizza A		
2	order pizza B		A 🚚
3	order pizza C		A 🚚, B 🚚
4	order pizza D	eat pizza A 🍕	B 🚚, C 🚚
5	order pizza E	eat pizza B 🍕	C 🚚, D 🚚

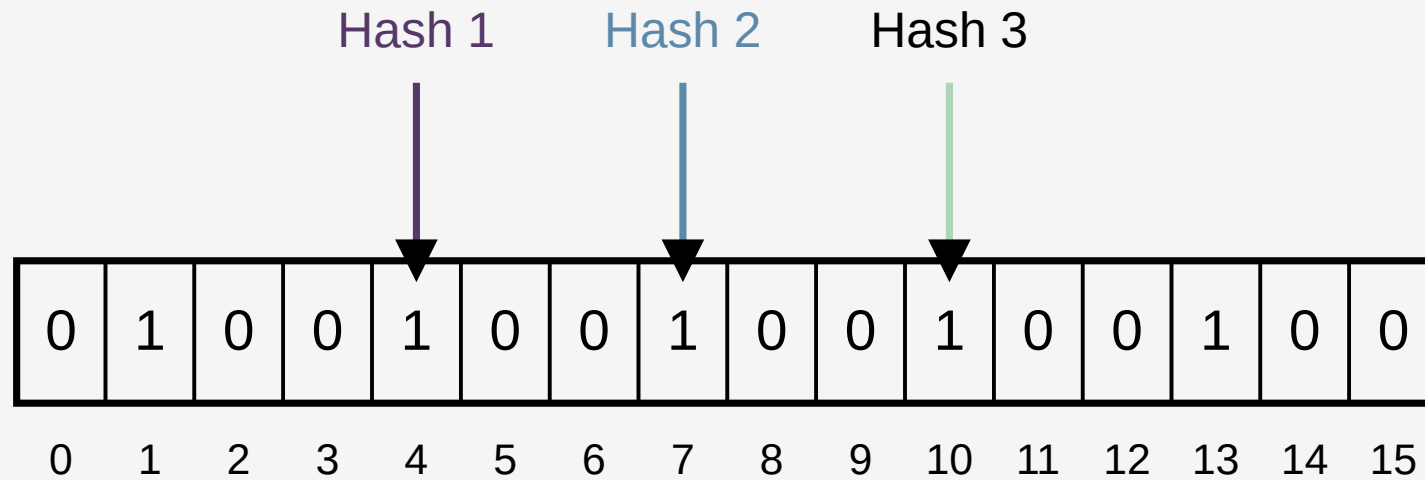
Little's Law

- Latency harms throughput
- Parallelism hides latency

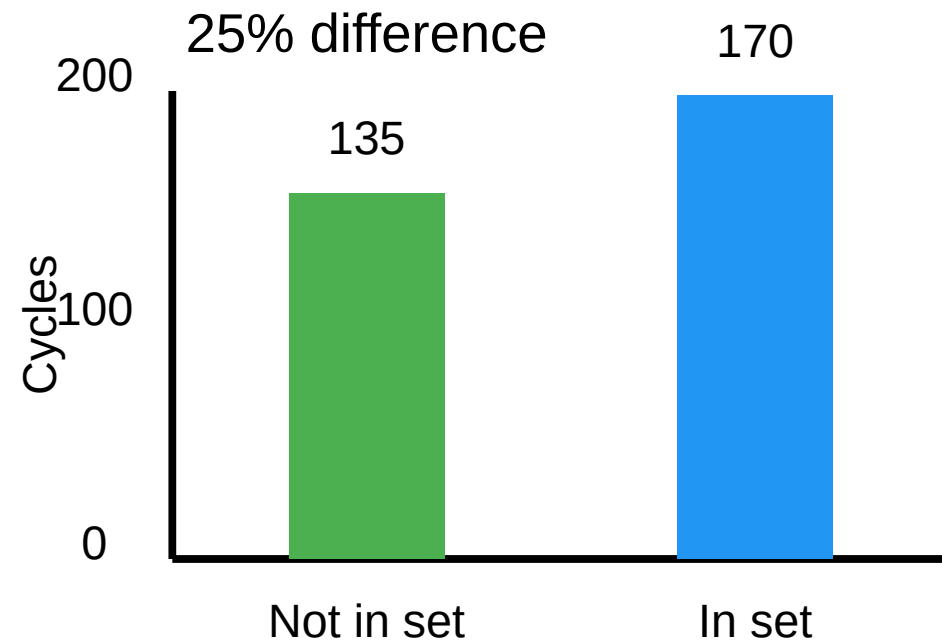
$$\text{throughput} = \frac{\text{parallelism}}{\text{latency}}$$



Bloom filter



Bloom filter (Intel Ice Lake, out-of-cache)



Same algorithm. Same hash functions. **Different memory access pattern.**

4. Data-level parallelism

The rest of this talk

Part 3

SIMD and SWAR

SIMD (Single Instruction, Multiple Data)

- Process 16, 32 or 64 bytes with **one** instruction
- Supported on every modern CPU — your phone included
- Not a niche feature: it is most of the silicon area of a modern core

The instruction sets

- **x64**: SSE2 (baseline, 128-bit), AVX2 (256-bit), AVX-512 (512-bit)
- **ARM**: NEON (128-bit, baseline on ARMv8), SVE/SVE2 (scalable)
- **RISC-V**: RVV (scalable)
- **POWER**: AltiVec/VSX · **LoongArch**: LSX, LASX
- **WebAssembly**: 128-bit SIMD, in every browser

Portability is mostly solved: compile several kernels, dispatch at runtime on CPU features. C++26 adds data-parallel types (`std::simd`).

SWAR: SIMD within a register

- Use plain 64-bit integer instructions
- A 64-bit register holds 8 bytes
- Fully portable C, C++, Rust, Go, Java...
- No intrinsics, no dispatch, no `#ifdef`
- Requires some cleverness

A great place to start.

Check whether we have 8 digits

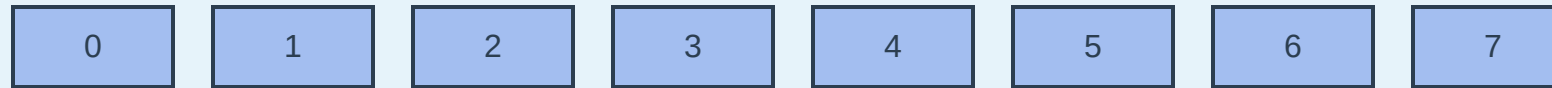
In ASCII/UTF-8, the digits 0, 1, ..., 9 have values 0x30, 0x31, ..., 0x39.

To recognize a digit:

- The high nibble should be 3.
- The high nibble should remain 3 if we add 6 (0x39 + 0x6 is 0x3f)

ASCII processing flow

ASCII numbers



ASCII codes (hex)



Zero lower 4 bits (AND 0xF0)



Add 6, shift right 4



OR result (hex)



```
// load 8 input bytes into val
bool is_made_of_eight_digits_fast(uint64_t val) noexcept {
    return !(((val + 0x4646464646464646)
              | (val - 0x3030303030303030))
            & 0x8080808080808080));
}
```

compiles to

```
add    rax, rdi
add    rdi, rdx
or     rax, rdi
test   rax, rdx
```

Four instructions for eight characters, and no branch.

Part 4

Case studies

Case study: parsing a number

- `1.3321321e-12` to `double`

```
double result;  
fast_float::from_chars(  
    input.data(), input.data() + input.size(), result);
```

- Used by major browsers (Safari, Chrome), GCC (12+), C#, Rust, MySQL, Go, Python
- About $4\times$ faster than the conventional alternatives

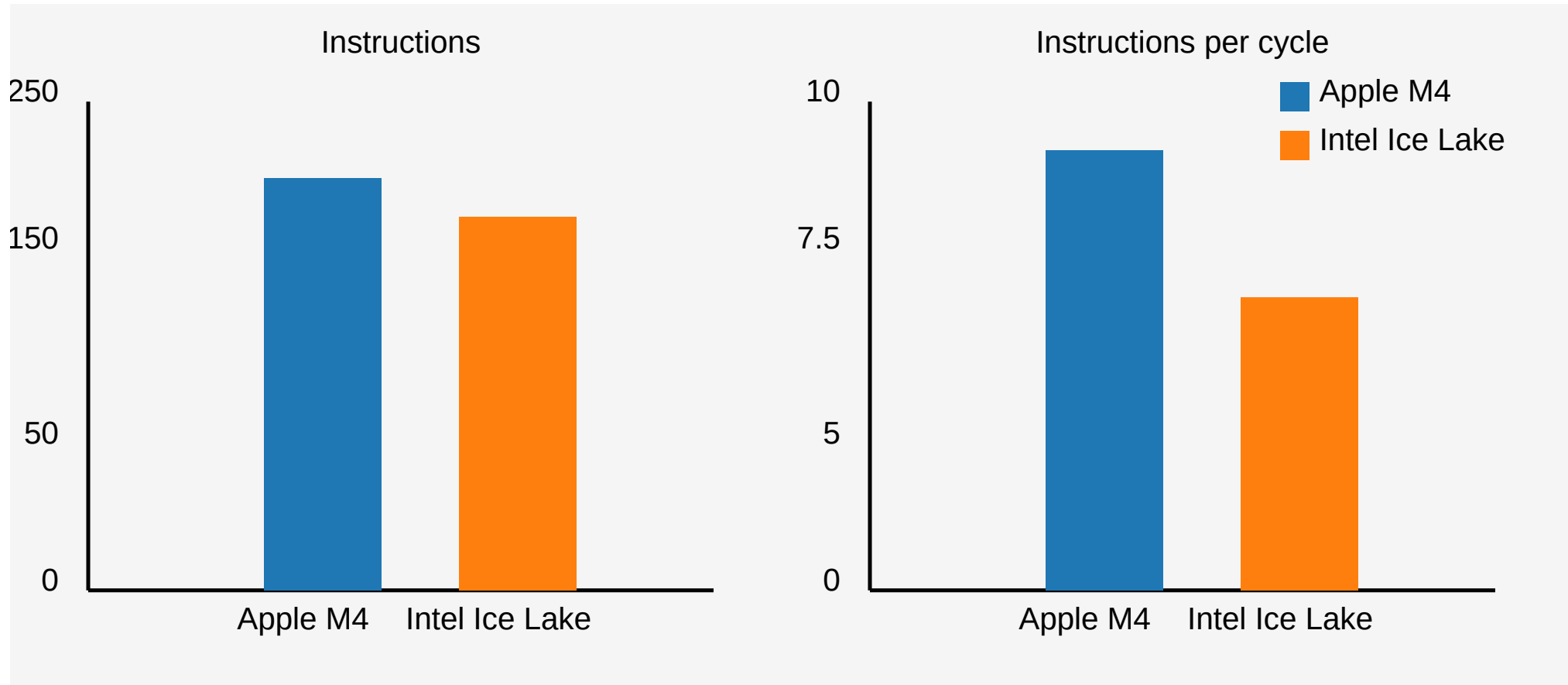
We massively reduced the number of CPU instructions required.

function	instructions
strtod	> 1000
our parser	≈ 200

Reference:

Number Parsing at a Gigabyte per Second, Software: Practice and Experience 51 (8), 2021

https://github.com/fastfloat/fast_float



The lesson

- We did not make the instructions faster.
- We **removed** instructions.
- Data parallelism is an instruction-count strategy.

Case study: 64 characters, 3 instructions

ASCII to lower case

```
For each character c
  If c - 'A' <= 'Z' - 'A' then
    c = c + 'a' - 'A'
  EndIf
EndFor
```

One byte per iteration. One unpredictable branch per byte.

64 characters in 3 instructions

- Compute $c - A$

```
__m512i ca = _mm512_sub_epi8(c, _mm512_set1_epi8('A'));
```

- Turn $c - 'A' \leq Z - A$ into a mask

```
__mmask64 is_upper = _mm512_cmple_epu8_mask(ca, _mm512_set1_epi8('Z' - 'A'));
```

- Add $a - A$ to c according to the mask

```
__m512i result = _mm512_mask_add_epi8(c, is_upper, c, to_lower);
```

No branch at all.

Let us check with llvm-mca

- `llvm-mca` is a static machine-code analyzer shipped with LLVM
- Feed it assembly, name a microarchitecture, get a cycle estimate
- No hardware required, no noise, no warm-up

```
llvm-mca -mcpu=icelake-server -iterations=100 kernel.s
```

The scalar loop: 1 byte per iteration

```
movzbl    (%rdi,%rax), %ecx
leal      -65(%rcx), %edx
cmpb      $26, %dl
jae        .LBB1_2
addb      $32, %cl
movb      %cl, (%rdi,%rax)
addq      $1, %rax
cmpq      %rax, %rsi
jne        .LBB1_1
```

The AVX-512 loop: 64 bytes per iteration

```
vmovdqu64    (%rdi,%rax), %zmm2
vpsubb       %zmm0, %zmm2, %zmm3
vpcmpub      $2, %zmm1, %zmm3, %k1
vpaddb       %zmm4, %zmm2, %zmm2 {%k1}
vmovdqu64    %zmm2, (%rdi,%rax)
addq         $64, %rax
cmpq         %rax, %rsi
jne          .LBB0_1
```

llvm-mca says...

	scalar	AVX-512
Iterations:	100	100
Instructions:	900	800
Total Cycles:	206	216
uOps Per Cycle:	4.37	4.63
IPC:	4.37	3.70

- Bytes processed: **100** vs **6400**
- Cycles per byte: **2.06** vs **0.034**

About 60× fewer cycles per byte.

Read that table again

- The scalar loop has the **higher IPC** (4.37 vs 3.70).
- The scalar loop retires **more instructions per cycle**.
- The scalar loop is **60× slower**.

IPC is not a performance metric. It is a diagnostic.

What llvm-mca will not tell you

- It assumes perfect branch prediction — the scalar loop's real cost is worse.
- It assumes every load hits L1 — real code will be memory bound.
- It models one microarchitecture at a time.

Use it to *explain* a measurement, not to replace one.

Case study: JSON

JSON

- Portable, simple, human-readable; used by ~97% of API requests
- Strings (escaped), numbers, objects, arrays
- Reading and writing JSON is often *slow*: 100 MB/s to 300 MB/s
- Slower than a fast disk or a fast network

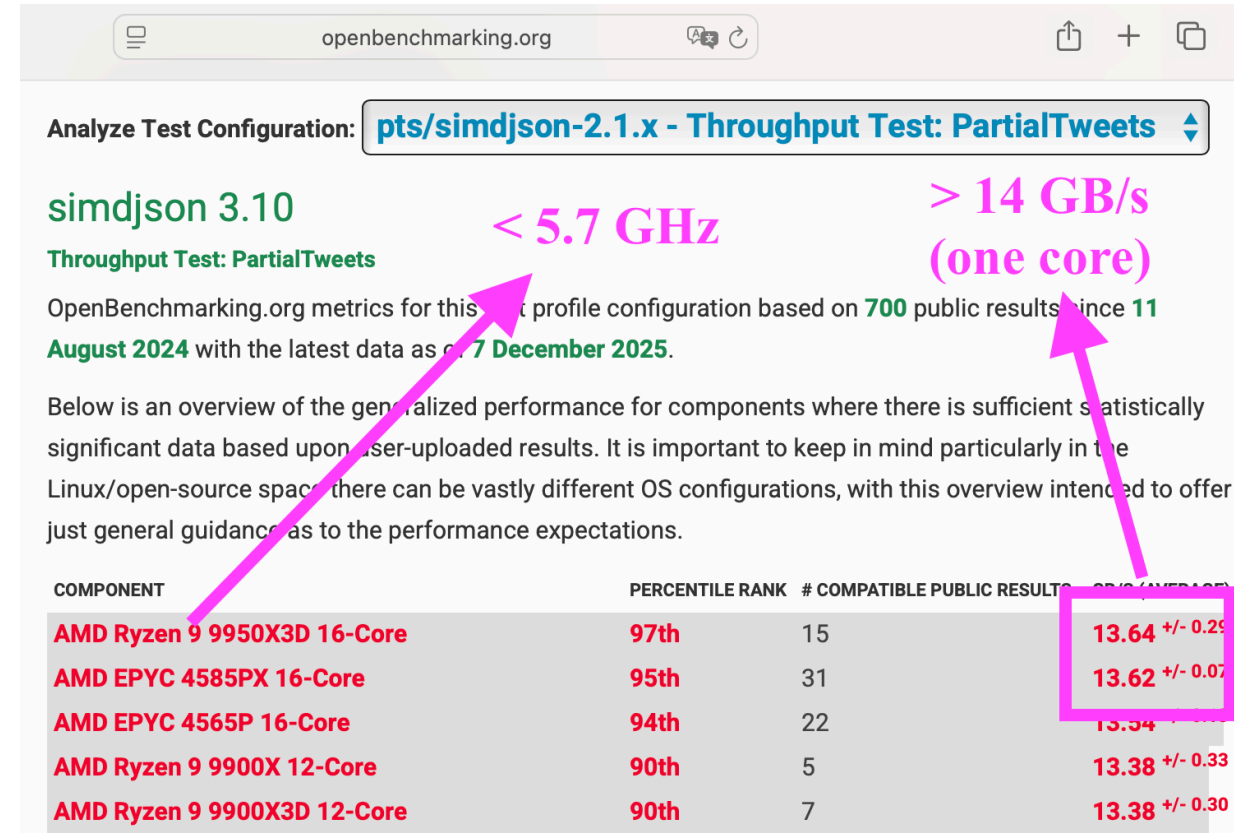
```
$ go run parse_twitter.go  
Parsed 0.63 GB in 6.961 seconds (90.72 MB/s)
```

"But JSON parsing is inherently serial"

- Every byte can change the meaning of every following byte.
- Strings contain escapes; escapes contain quotes.
- It is a textbook state machine.

This was the conventional wisdom. It was wrong.

- openbenchmarking.org
- 14 GB/s, at less than 5.7 GHz
- Parsing JSON at better than **2.5 bytes per cycle**



You are probably using simdjson

- Node.js, Bun, Deno, Electron
- ClickHouse
- WatermelonDB, Apache Doris, Meta Velox, Milvus, QuestDB, StarRocks



```
{
  "nom": "Marie Dubois",
  "age": 28,
  "ville": "Paris",
  "loisirs": ["lecture", "voyage", "yoga"],
  "contact": {
    "email": "marie.dubois@example.com",
    "phone": "+33123456789"
  }
}
```

Every major JavaScript engine parses JSON with SIMD

- `JSON.parse` in Node.js, Bun and Deno is data-parallel.
- Billions of calls per second worldwide.
- Nobody had to change a single line of JavaScript.

simdjson: two-stage design

Stage 1 (data-parallel): scan the whole document with SIMD

- find every structural character and the start of every string
- validate UTF-8
- produce an index

Stage 2 (mostly serial): walk the index and build values

- the hard, branchy work now runs on ~5% of the bytes

simdjson: the numbers

- Structural scan: about **10 GB/s**
- UTF-8 validation: about **30 GB/s**
- Minification: **10 to 20 GB/s**
- Fast skipping: only parse what you actually read

Classifying characters

We need to sort every byte into a class:

- comma (0x2c) ,
- colon (0x3a) :
- brackets (0x5b, 0x5d, 0x7b, 0x7d): [,], {, }
- white-space (0x09, 0x0a, 0x0d, 0x20)
- everything else

A switch statement per byte? No.

Vectorized classification

- Most SIMD instruction sets support 'vectorized lookup tables' (at least 16-element)
- With a 256-element table we could just compute `H(c)`
- With 16-element tables, we need two tables `H1` and `H2`
- Find `H1` and `H2` such that the bitwise AND of the lookups classifies the character:

```
H1(c & 0xf) & H2(c >> 4)
```

```
low_nibble_mask  = {16, 0, 0, 0, 0, 0, 0, 0, 0, 8, 12, 1, 2, 9, 0, 0};  
high_nibble_mask = {8, 0, 18, 4, 0, 1, 0, 1, 0, 0, 0, 3, 2, 1, 0, 0};
```

Five instructions, 16 to 64 bytes at a time:

```
nib_lo  = input & 0xf;  
nib_hi  = input >> 4;  
shuf_lo = lookup(low_nibble_mask, nib_lo);  
shuf_hi = lookup(high_nibble_mask, nib_hi);  
return shuf_lo & shuf_hi;
```

This trick generalizes: any 256-way classification into 8 classes.

Serialization is also a data-parallel problem

- JSON requires escaping `"`, `\`, and control characters.
- Almost no string actually needs escaping.
- So: check 16 to 64 bytes at once, and take the fast path.

SIMD string escaping

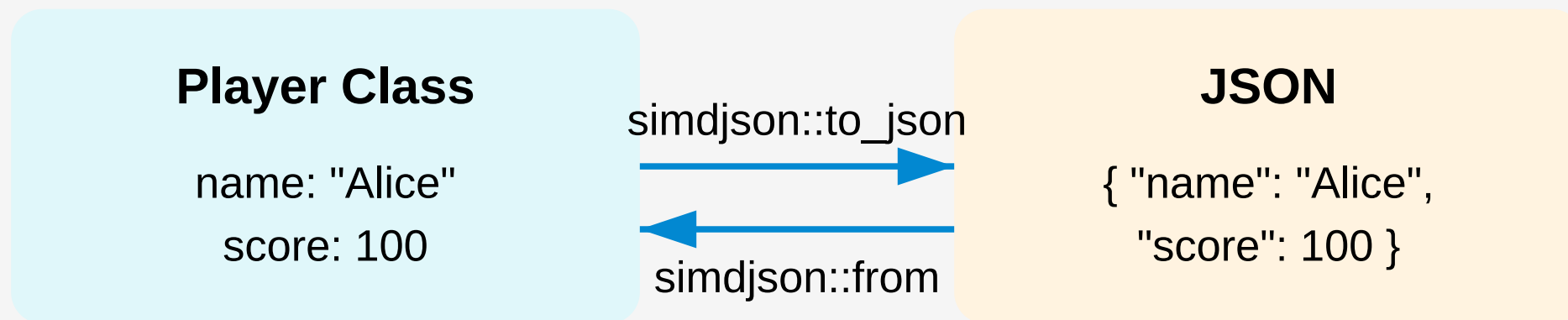
Traditional (1 byte at a time):

```
for (char c : str) {  
    if (c == '"' || c == '\\\' || c < 0x20)  
        return true;  
}
```

SIMD (64 bytes at once):

```
auto chunk = load_64_bytes(str);  
auto needs_escape = check_all_conditions_parallel(chunk);  
if (!needs_escape)  
    return false; // Fast path!
```

C++26 compile-time reflection

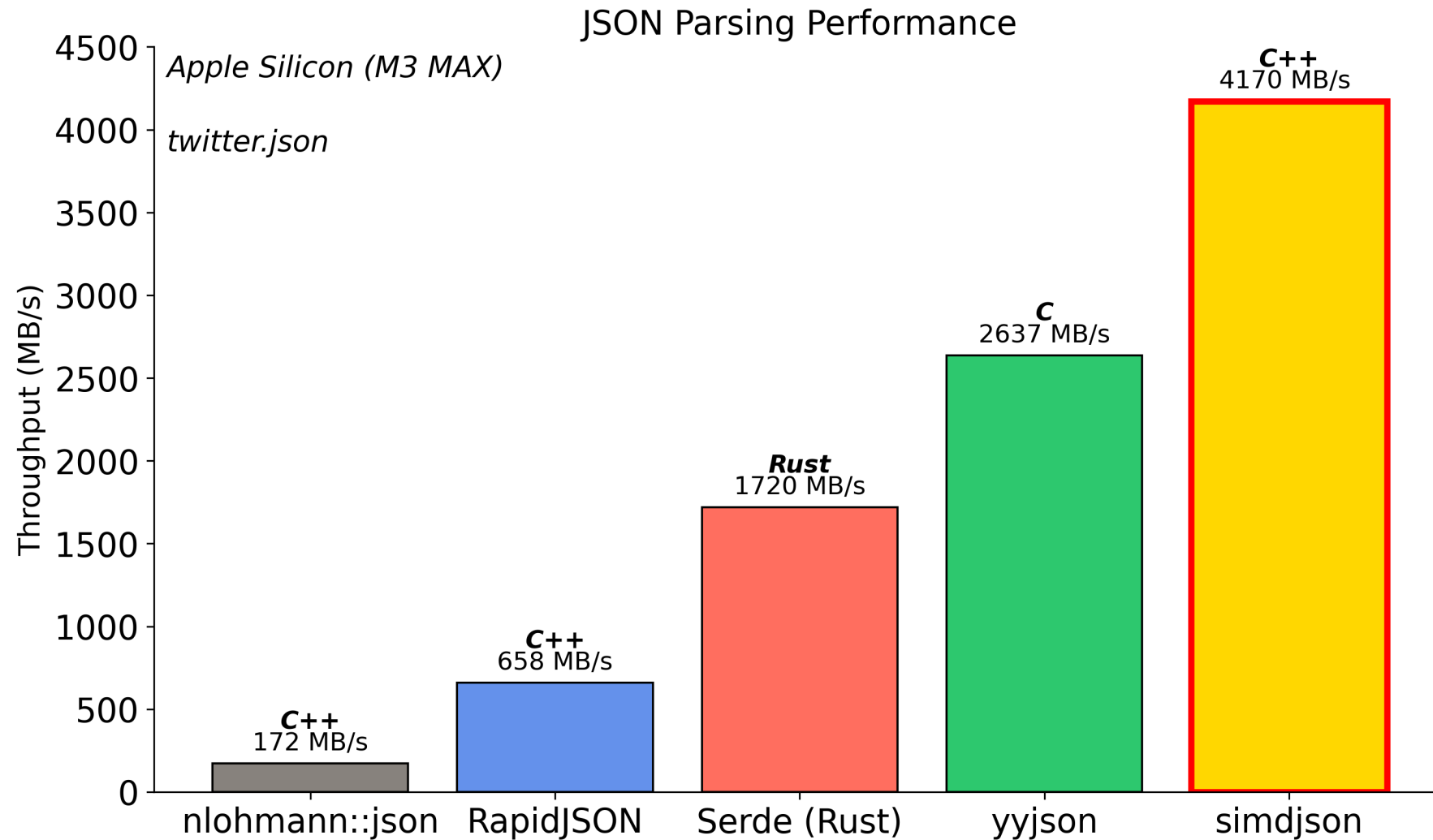


One line each way (C++26)

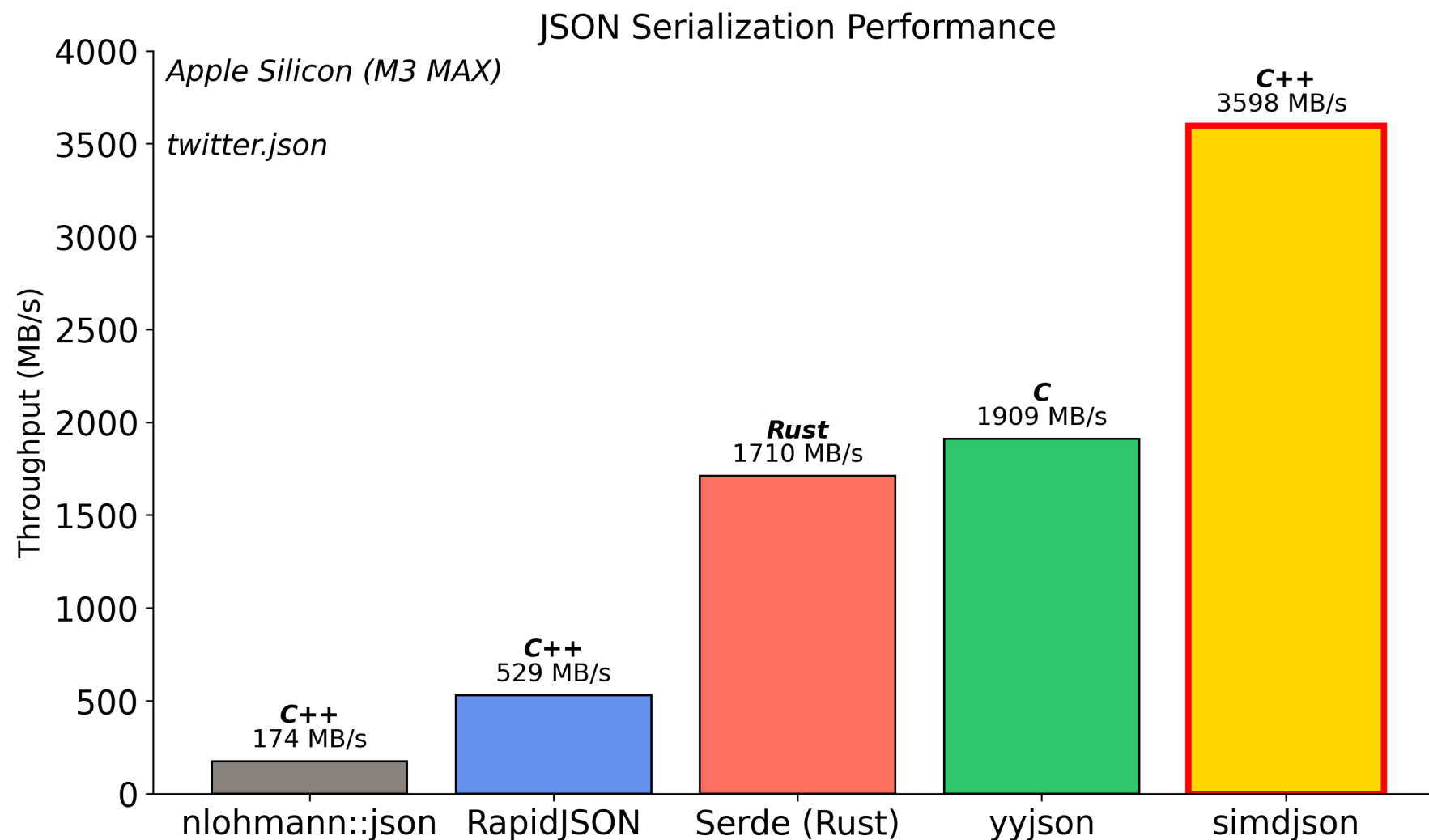
```
struct Player {  
    std::string username;  
    int level;  
};  
  
Player load_player(std::string& json_str) {  
    return simdjson::from(json_str);  
}  
  
std::string save_player(const Player& p) {  
    return simdjson::to_json(p);  
}
```

No macros. No code generation step. No runtime reflection cost.

Deserialization (Apple Silicon)



Serialization (Apple Silicon)



Case study: Unicode

simdutf

- Inside Safari, Chrome, Node.js, Bun
- Unicode transcoding and validation at gigabytes per second
- Base64 too
- x64, ARM, POWER, RISC-V, LoongArch



simdutf

Unicode (UTF-16)

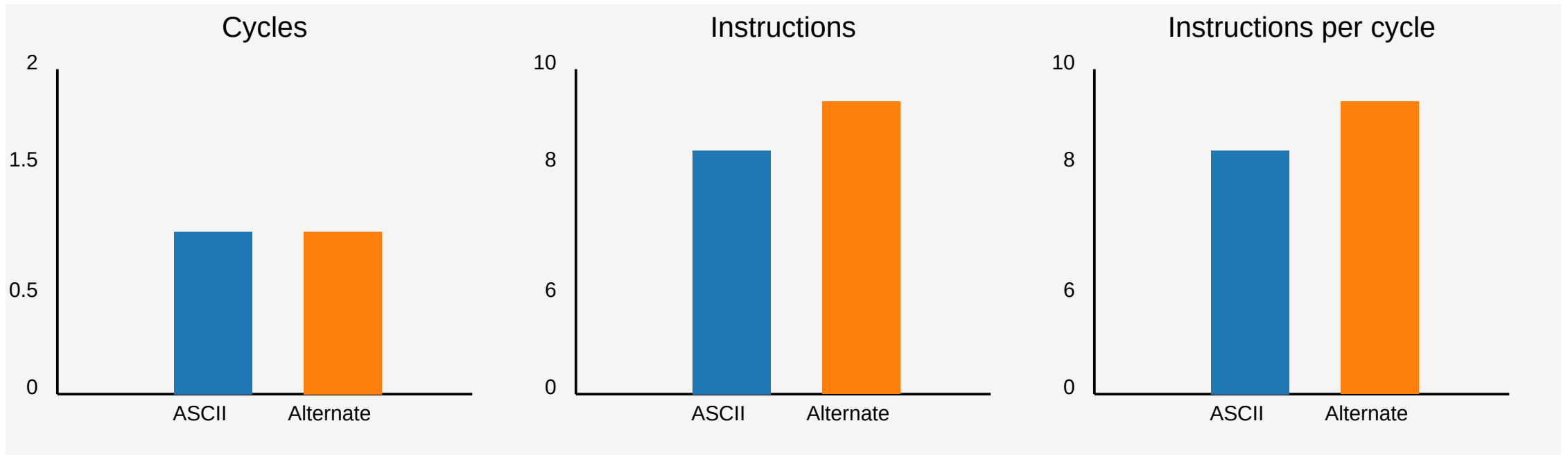
- Code points from U+0000 to U+FFFF: a single 16-bit value.
- Beyond: a *surrogate pair*, U+D800 – U+DBFF followed by U+DC00 – U+DFFF .
- A lone surrogate makes the string ill-formed.

Every JavaScript string, every Java string, every Windows filename.

Validate

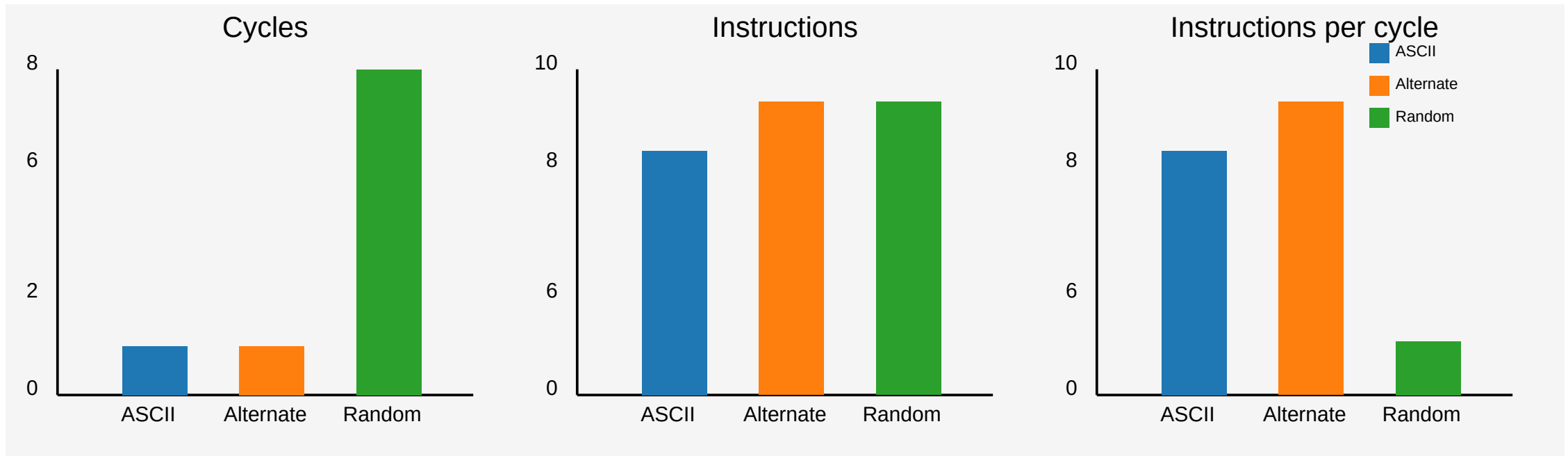
```
PROCEDURE validate_utf16(code_units)
  i ← 0
  WHILE i < |code_units|
    unit ← code_units[i]
    IF unit ≤ 0xD7FF OR unit ≥ 0xE000 THEN
      INCREMENT i
      CONTINUE
    IF unit ≥ 0xD800 AND unit ≤ 0xDBFF THEN
      IF i + 1 ≥ |code_units| THEN
        RETURN false
      next_unit ← code_units[i + 1]
      IF next_unit < 0xDC00 OR next_unit > 0xDFFF THEN
        RETURN false
      i ← i + 2 // Valid surrogate pair
      CONTINUE
    RETURN false
  RETURN true
```

Performance results (Apple M4)



1 character per cycle might be just 4 GB/s — slower than your disk.

Now make the input adversarial



We are now barely at 1 GB/s. **The branch predictor was doing the work.**

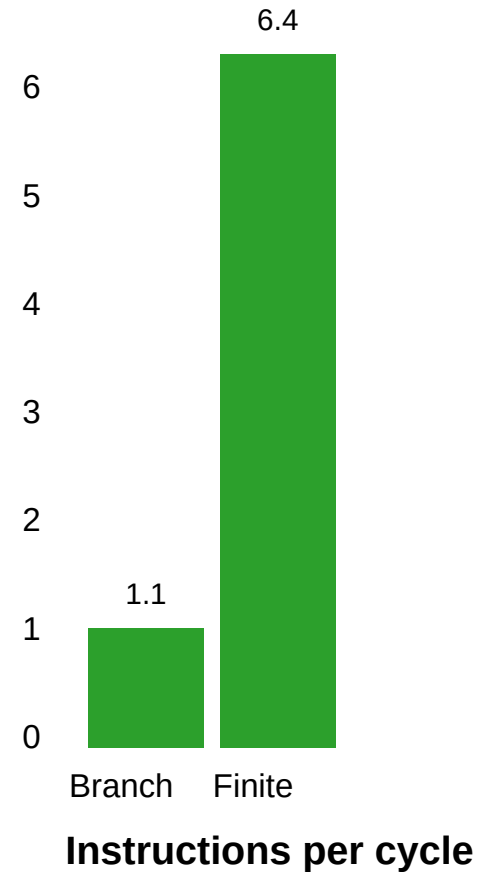
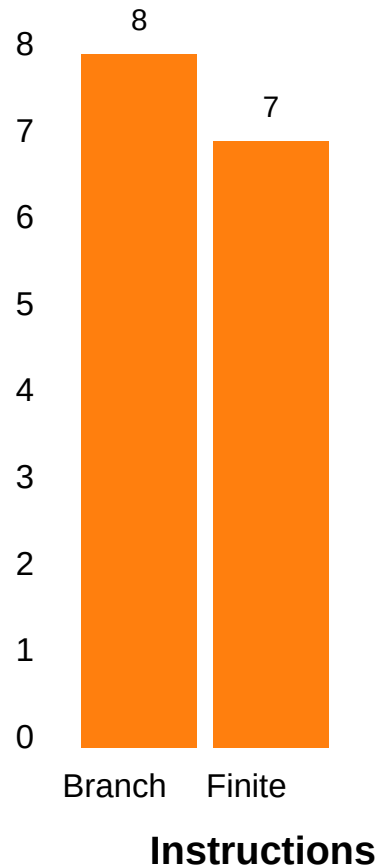
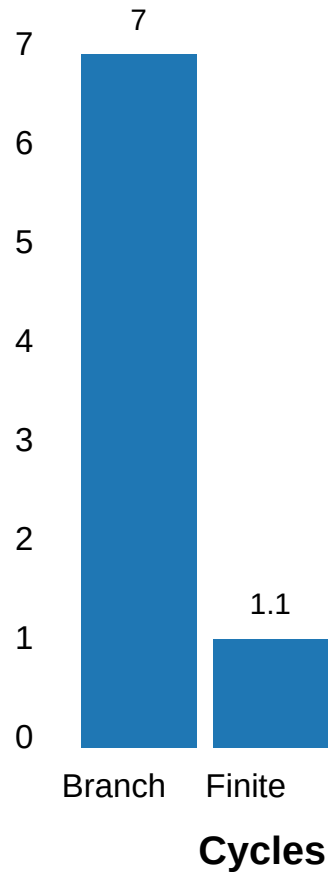
Finite state machine to the rescue

```
static uint8_t transition_table[3][256] = { {...}, {...}, {...} };

bool is_valid_utf16_ff(std::span<uint16_t> code_units) {
    uint8_t state = 0; // Start in Initial state
    for (auto code_unit : code_units) {
        uint8_t high_byte = code_unit >> 8;
        state = transition_table[state][high_byte];
    }
    return state == 0; // Valid only if we end in Initial state
}
```

Three states: default, just saw a high surrogate, error. **No branches.**

The finite-state approach can be $7\times$ faster

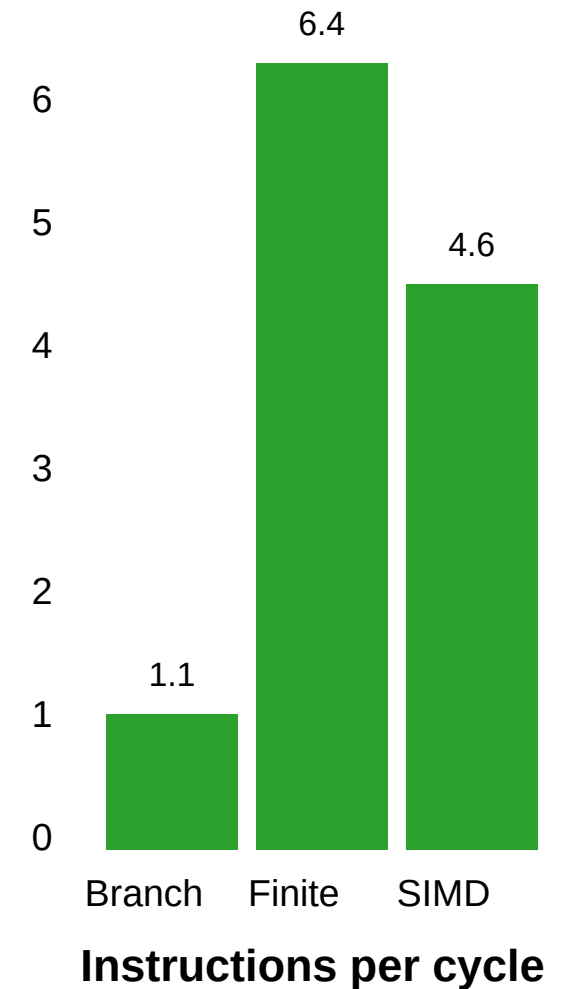
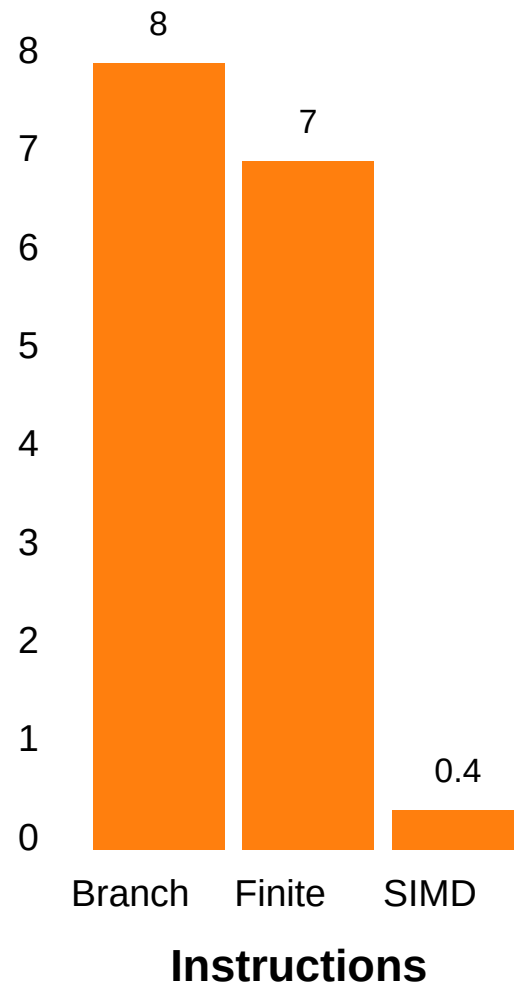
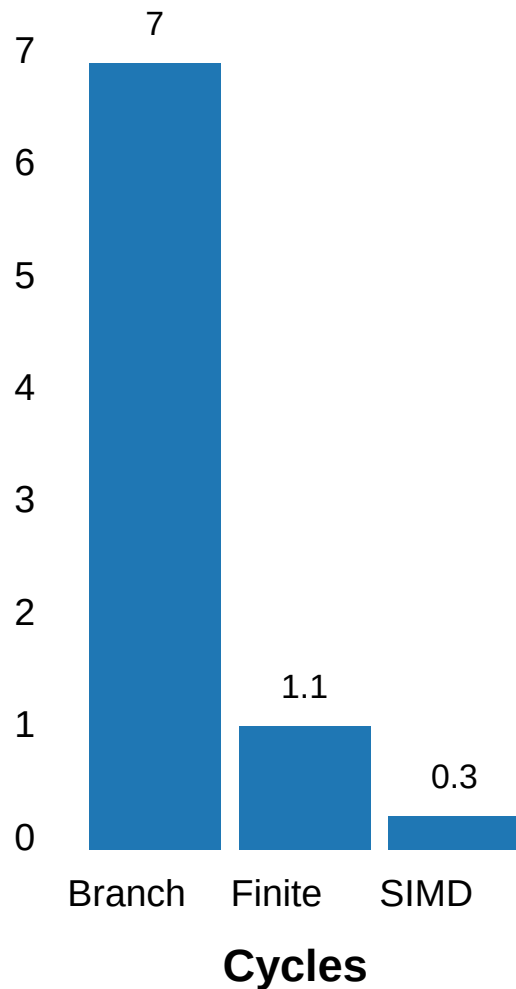


But we can do better than validation

```
const str = "ab\uD800";  
console.log(str.toWellFormed());  
// "ab?"
```

`String.prototype.toWellFormed()` must *copy and repair*, not merely check.

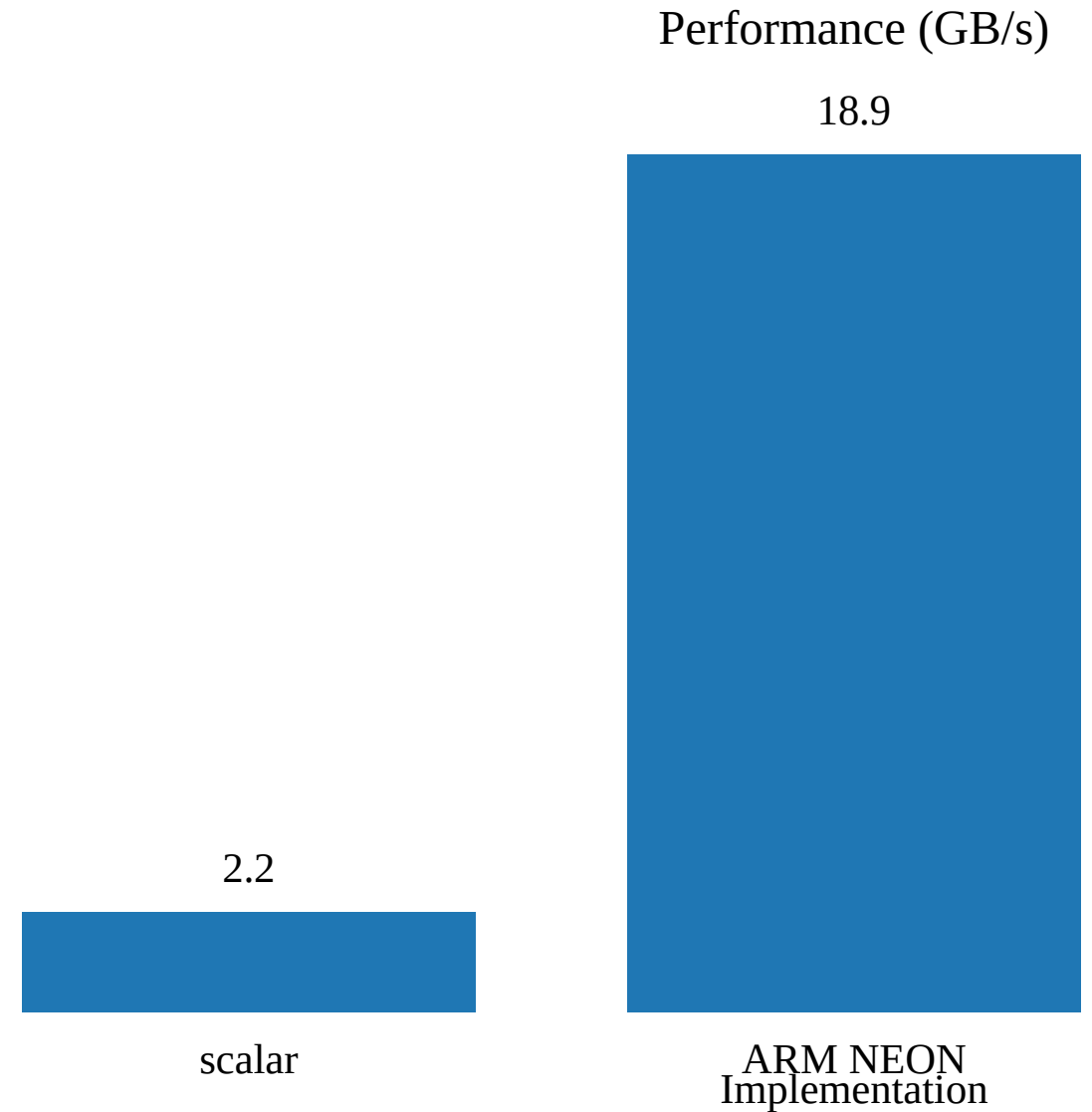
UTF-16, random (adversarial), Apple M4



UTF-16 correction, Apple M4

	scalar	ARM NEON
GB/s	2.2	18.9
ins/byte	12.0	0.9

13× fewer instructions per byte.



In the browser (Apple M4)

- Chromium: 16 GB/s (**uses our new function**)
- Firefox: 3.4 GB/s
- Safari: 1.2 GB/s

Test it yourself: <https://lemire.github.io/browserwellformed/>

Case study: base64

Base64

- Encodes binary data as text using 64 characters (A-Z, a-z, 0-9, +, /)
- 3 bytes input → 4 characters output (33% overhead)
- Data URLs, email, JWTs, web APIs, embedded images
- "Hello, world!" → SGVsbG8sIFdvcmxkIQ==

Bit manipulation on a fixed schedule: **the ideal SIMD problem.**

New JavaScript functions

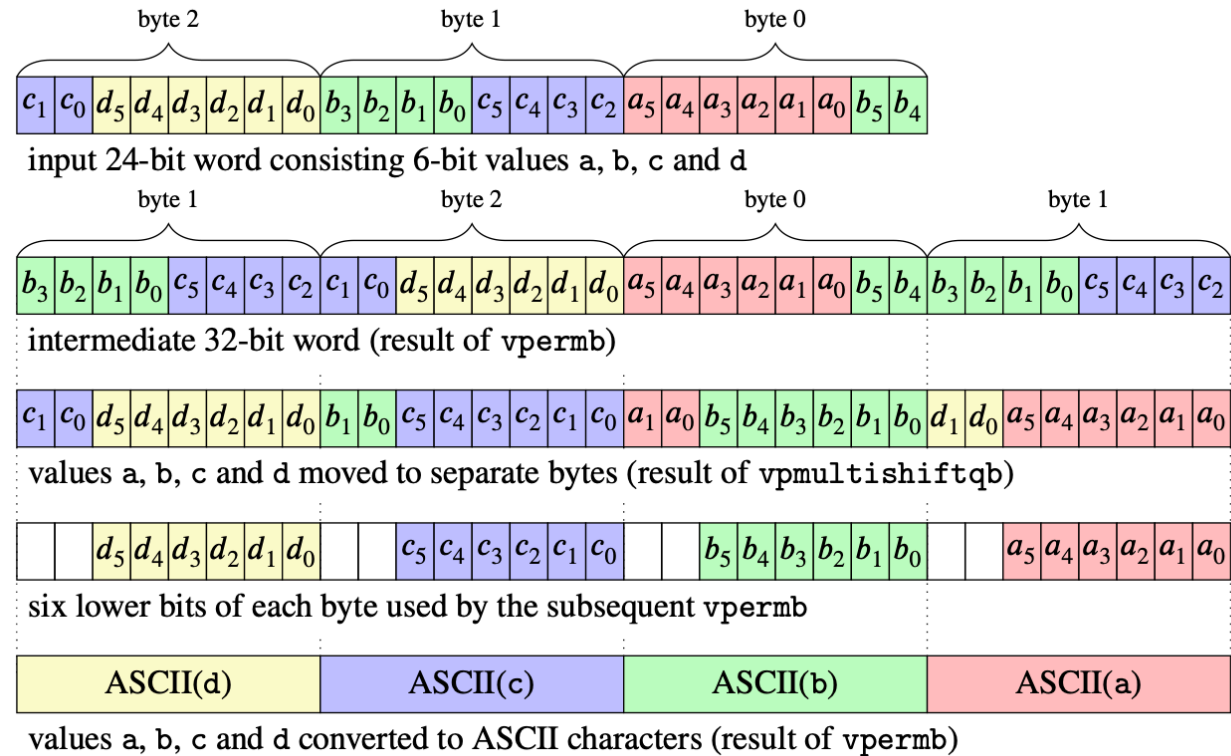
```
const b64 = Uint8Array.prototype.toBase64(bytes);  
const recovered = Uint8Array.fromBase64(b64);
```

function (Safari, Apple M4)	speed
<code>Uint8Array.fromBase64()</code>	11 GiB/s
<code>Uint8Array.toBase64()</code>	20 GiB/s

Test in your browser: <https://simdutf.github.io/browserbase64/>

AVX-512 base64 encoding

- Encoding a 64-byte block requires only **two** non-memory instructions:
`vpermb` (twice) and `vpmultishiftqb`.
- The right instruction turns an algorithm into a lookup.

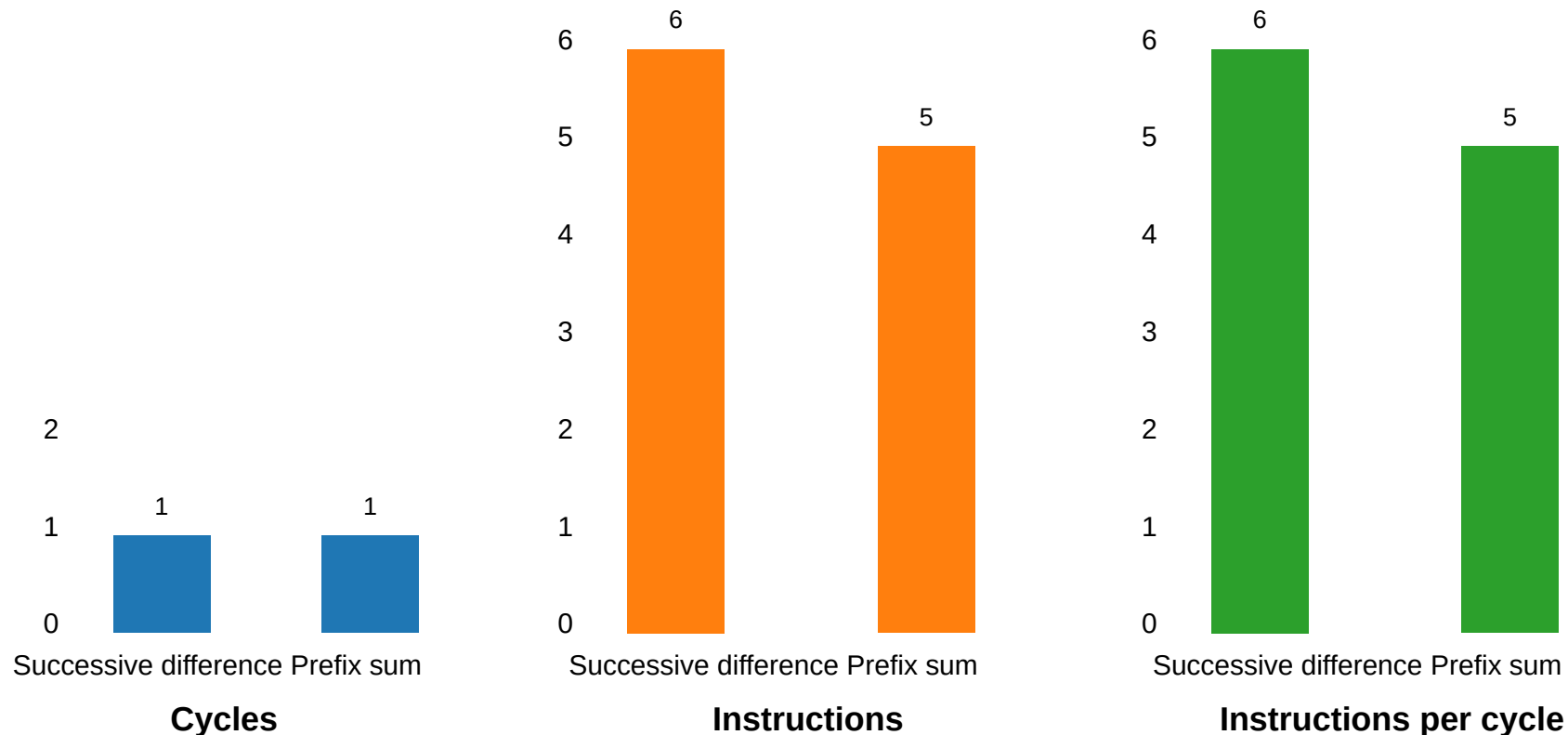


Part 5

Why your compiler will not do this for you

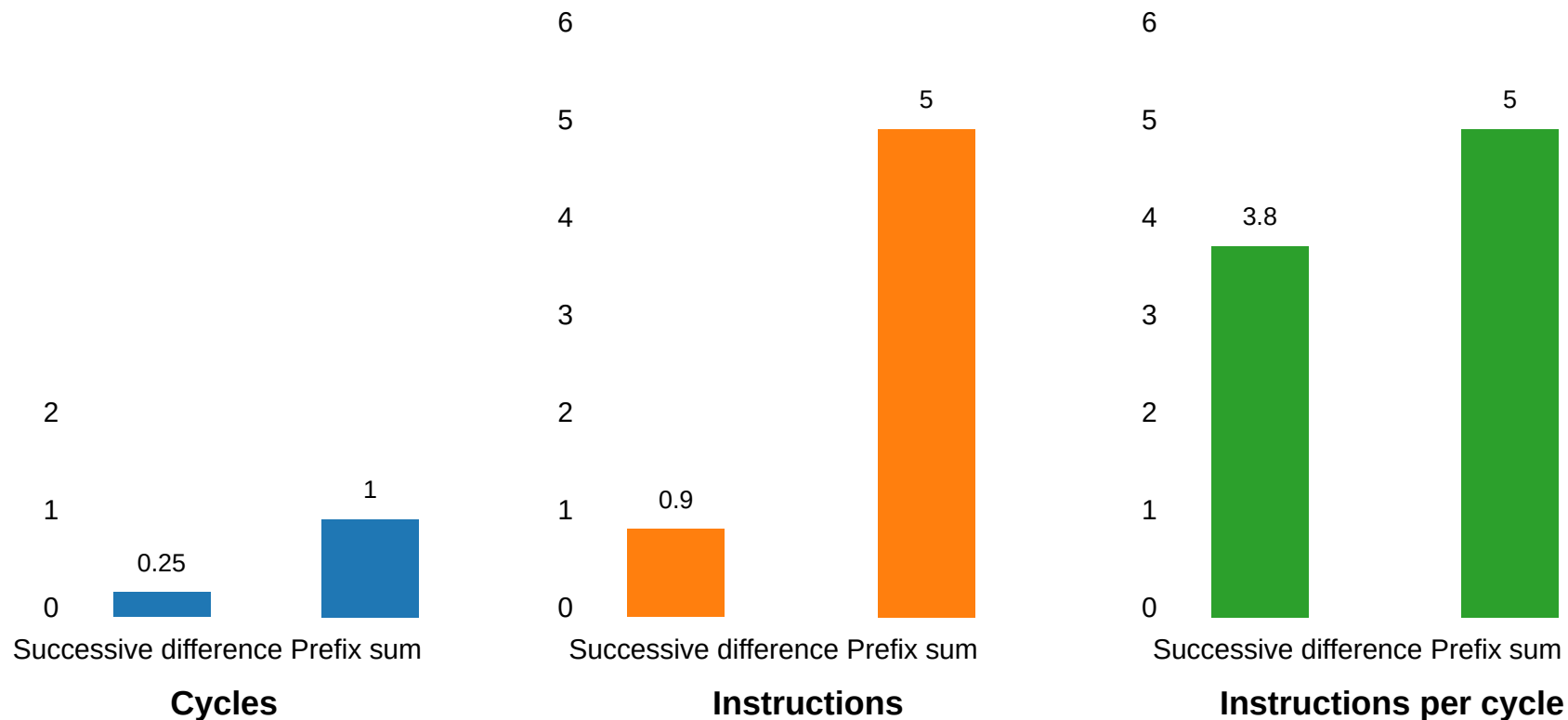
Auto-vectorization: the good case

Successive differences: `out[i] = in[i] - in[i-1]`



Scalar: **1 cycle** per element, 6 instructions per element.

Auto-vectorization: the good case



Vectorized: **0.25 cycles** per element, 0.9 instructions per element. 4× faster, for free.

Note the instructions per cycle went **down**, from 6 to 3.8.

But look at the prefix sum

- `out[i] = out[i-1] + in[i]`
- Same shape of loop. Same data. Same compiler.
- **Unchanged:** 1 cycle per element, in both builds.
- There is a loop-carried dependency, so the compiler gives up.

And yet the prefix sum *does* vectorize

- Shift-and-add, $\log_2 n$ steps: a classic parallel-prefix network.
- The compiler cannot invent it, because it is **a different algorithm**, not a different schedule.

This is the whole talk in one slide.

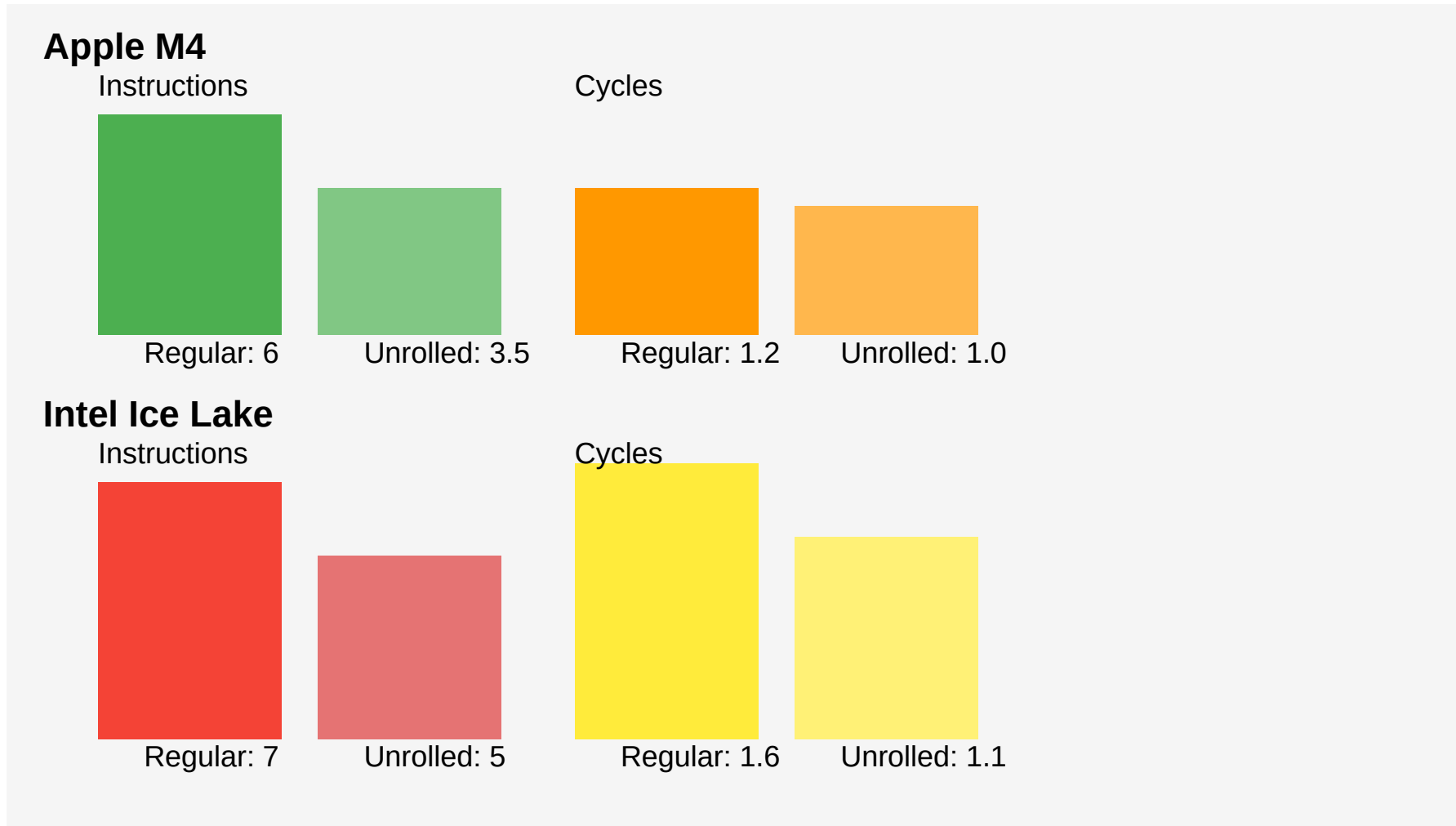
What compilers can do

- Unroll loops
- Vectorize simple, dependency-free, contiguous loops
- Choose instruction schedules

What compilers cannot do

- Change your data layout
- Change your algorithm
- Decide that a rare case can be handled on a slow path

Unrolling is not vectorizing



Fewer cycles, yes. But you are still touching one element at a time.

Part 6

Can an LLM write your SIMD code?

The honest answer: partly

Frontier models in 2026 are genuinely good at:

- Recalling intrinsic names and semantics (better than I am)
- Translating a working NEON kernel to AVX2, or to RVV
- Writing the scalar reference implementation and the test harness
- Explaining an unfamiliar instruction

Where they still struggle

- **Inventing** the vectorized algorithm (the nibble-lookup trick, the prefix network)
- Reasoning about throughput vs. latency and port pressure
- Knowing that a 512-bit instruction may downclock the core
- Noticing that the fast path is fast only on *your* data

They optimize what you measured. They do not choose what to measure.

What actually works: close the loop

Give the agent the three things it cannot produce on its own:

1. **A benchmark** it can run, over realistic inputs
2. **A differential fuzzer** against a scalar reference
3. `llvm-mca` (or `perf`) so it can see cycles, not vibes

Then let it iterate.

A workable prompt shape

Here is a scalar reference implementation and a fuzzer that compares any candidate against it.

Here is a benchmark: ``make bench`` prints cycles per byte.

Write an AVX2 version. After each attempt, run the fuzzer and the benchmark, and run `llvm-mca` on the inner loop. Do not stop until the fuzzer passes and cycles/byte is below 0.2.

The constraint is the contribution. The agent supplies the patience.

The failure mode to watch for

- An agent will happily produce SIMD code that is **correct on your test inputs** and wrong on the surrogate pair, the truncated block, the empty string, the 63-byte tail.
- Tail handling is where hand-written SIMD goes to die, and LLMs inherit that.
- **Fuzz the tails. Fuzz the alignment. Fuzz the adversarial input.**

The division of labour

you	the agent
choose the algorithm	write the intrinsics
define correctness	run the fuzzer
define the benchmark	iterate on the schedule
decide when it is fast enough	port to the other instruction sets

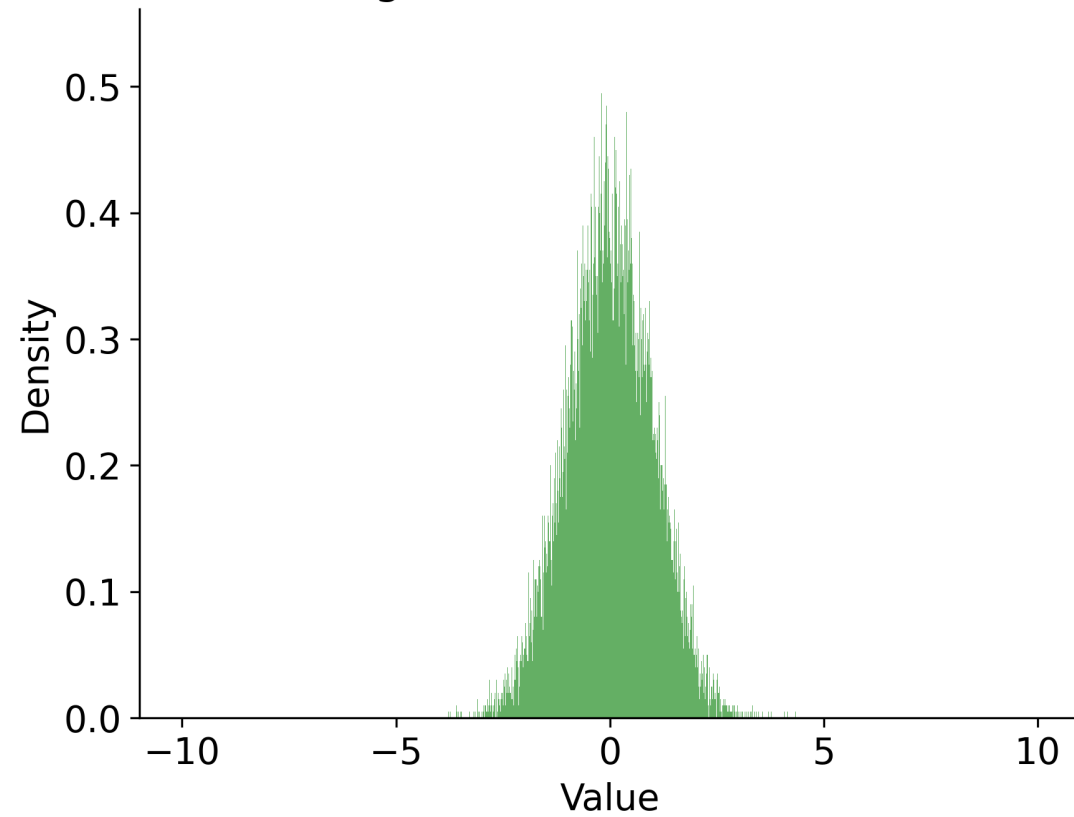
Part 7

Measure properly, or do not bother

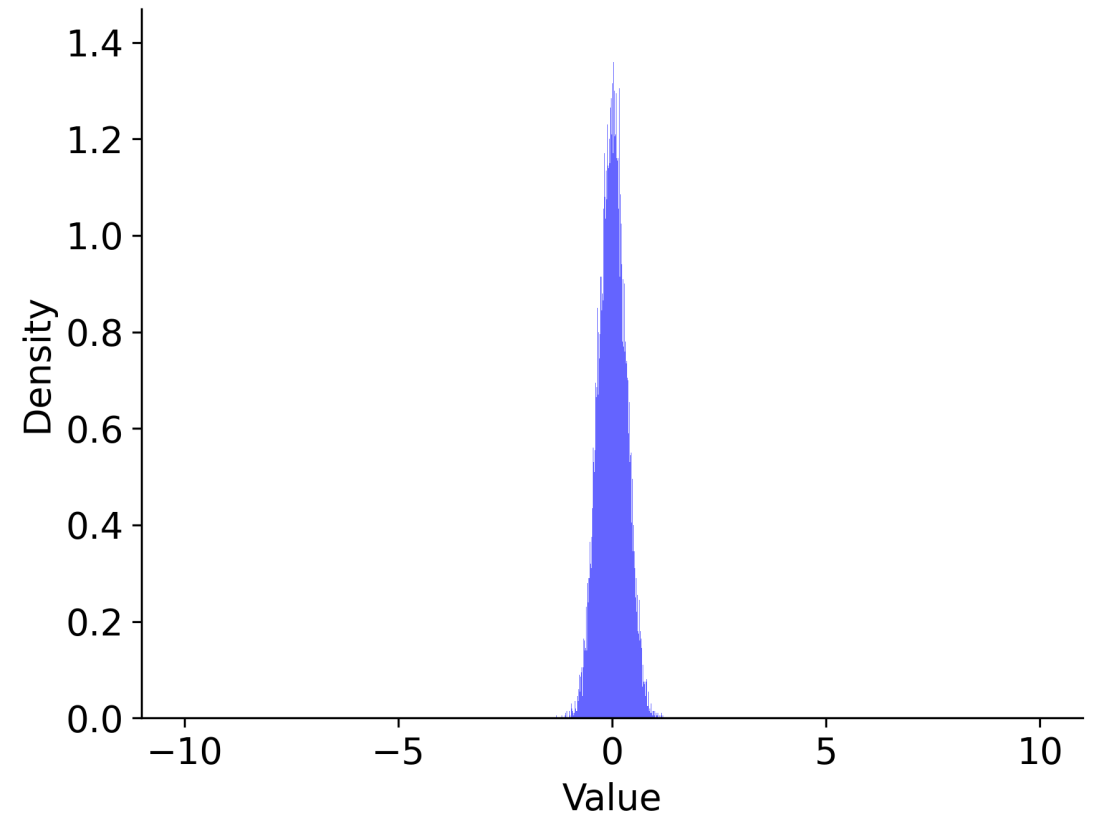
Measurements

- We often assume that measurements (timings) are normally distributed.
- If they were, the 'error' would fall off as $1/\sqrt{N}$.
- It is often an incorrect assumption.

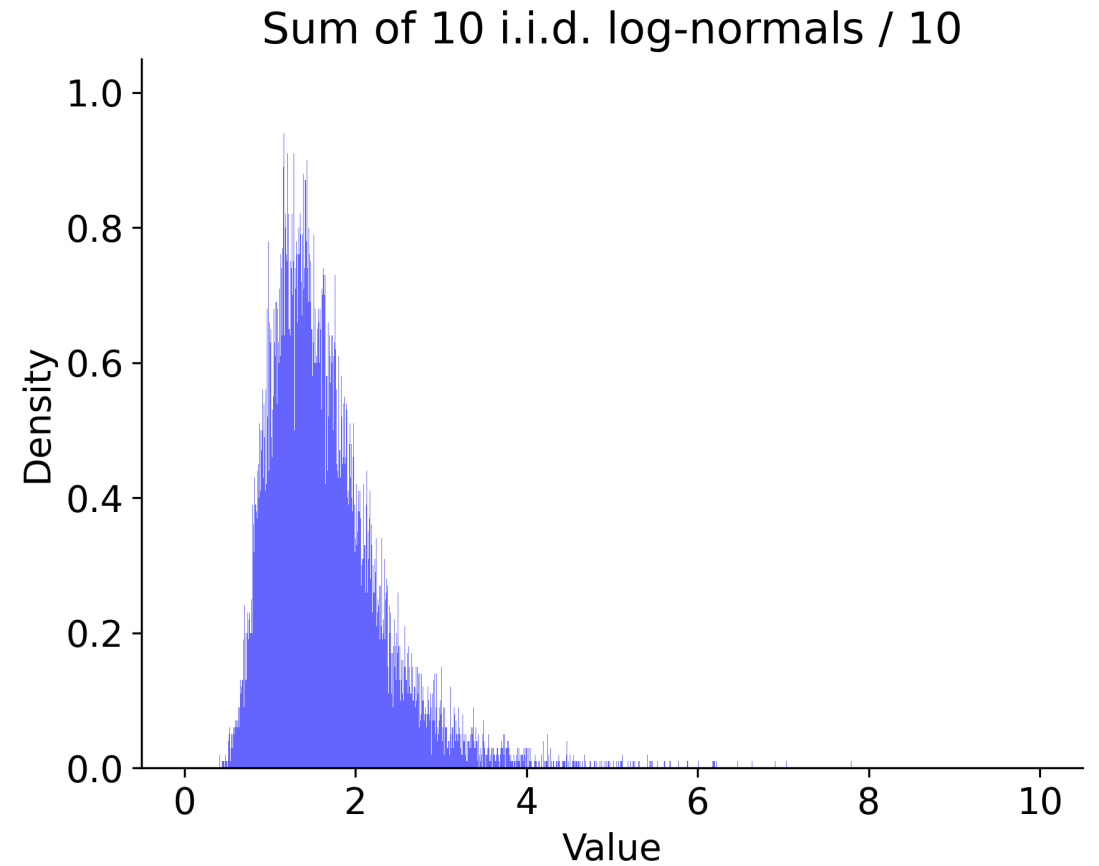
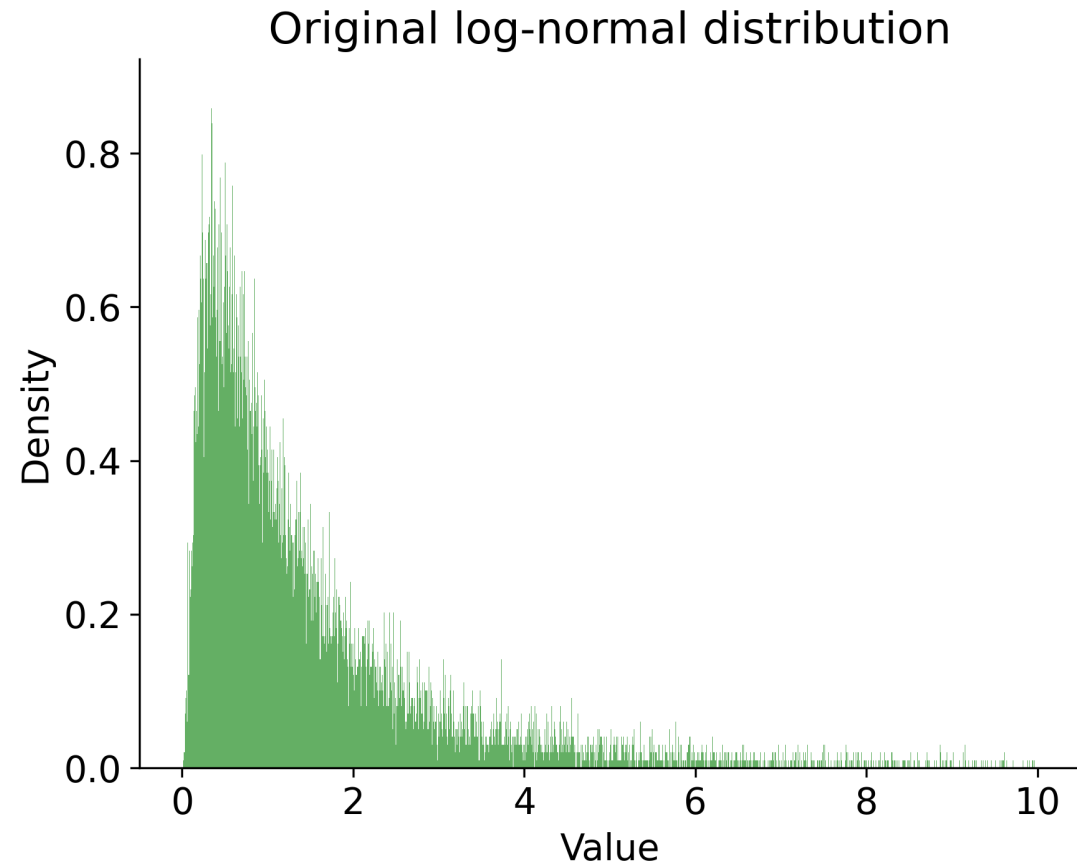
Original normal distribution



Sum of 10 i.i.d. normals / 10



What if we dealt with log-normal distributions?



Real-world measurements

- You cannot assume normality
- Measurements are **not independent**
- Reality: the absolute **minimum** is often the *reliable* metric
- Margin: the difference between the mean and the minimum

Use performance counters

Timings tell you *that* something is slow. Counters tell you *why*.

- **instructions retired** — did I actually remove work?
- **cycles** — the ground truth
- **branch misses** — is the predictor carrying me?
- **cache misses** — am I memory bound after all?

`perf stat`, `Instruments`, or a library such as `performancecounters`.

The diagnostic table I always build

variant	ns/byte	instr/byte	cycles/byte	IPC	branch miss/byte
scalar					
SWAR					
SIMD					

If instructions/byte did not drop, you did not do data parallelism.

If cycles/byte did not drop, find out which counter did move.

Hot-spot engineering fails

- A profiler shows you where the cycles are *now*.
- It does not show you the 30% of instructions spread evenly over every function.
- Reducing the total instruction count beats optimizing the top-of-profile function.

When is data parallelism worth it?

Good signs

- You touch every byte or every element
- The work per element is small and uniform
- You have an unpredictable branch per element
- You are validating, scanning, transcoding, filtering, or counting

Bad signs

- Deep pointer chasing
- Genuinely irregular control flow with expensive bodies
- You are already memory bound at full bandwidth

It applies more often than you think

- JSON parsing — every JavaScript engine
- Unicode validation — every browser
- base64 — the JavaScript standard library
- Number parsing — every compiler toolchain
- Bitmap indexes, hashing, compression, `memchr`, CSV, regex prefiltering

"Inherently serial" is usually a statement about the algorithm you happen to know.

What to take home

1. Clock speeds are flat; parallelism is where performance lives.
2. Data parallelism reduces **instructions**, not just time.
3. Your compiler schedules; it does not redesign. That part is yours.
4. Branchy code looks great in a synthetic benchmark and dies on real data.
5. IPC is a diagnostic, not a goal. The minimum time is your metric.
6. AI agents are excellent hands and mediocre architects — give them a benchmark and a fuzzer.

Interested? Check these projects

- **simdjson** — the fastest JSON parser in the world <https://simdjson.org>
 - Node.js, Bun, Deno, Electron
 - ClickHouse, WatermelonDB, Apache Doris, Meta Velox, Milvus, QuestDB, StarRocks
- **simdutf** — Unicode (UTF-8/16/32) and base64 <https://github.com/simdutf/simdutf>
 - Node.js, Bun, WebKit (Safari), Chromium (Chrome, Edge)
- **fast_float** — number parsing https://github.com/fastfloat/fast_float
- **Roaring Bitmaps** — <https://roaringbitmap.org>

Credit

- simdjson reflection work with Francisco Geiman Thiesen (Microsoft)
- simdutf UTF-16 correction is joint work with Robert Clausecker
- simdjson and simdutf are community efforts (Geoff Langdale, John Keiser, Paul Dreik, Yagiz Nizipli and others)

Questions?

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